ARITHMETIC AND GEOMETRY OF AN M_{23} HURWITZ SCHEME

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Abstract. Huang, Jackson, Lee, Poonen, Pries, and Zhang constructed a regular M_{23} Galois cover from a seven-element inner Nielsen class. We characterize its distinguished point by canonical geometry. On each genus-4 quotient curve, the osculating planes at the two totally ramified points meet in a line. This line meets exactly one of the seven curves in two reduced points and misses the other six. The condition is invariant under Galois conjugation and exchange of the marks, so uniqueness gives a global Galois-fixedness argument. We also construct a lift with this incidence from explicit characteristic-23 tail covers and a logarithmic deformation datum. Marked tame descent and the effective ramification divisor determine its canonical reduction. Comparing the two tail actions on the same 23 sheets selects the required deformation; Hensel uniqueness and an independently verified exact model prove incidence in characteristic zero. This computer-assisted construction uses more than the harmonic position of the reduced marked points. The exact Hurwitz algebra is $K_0 \times L$, where $K_0 = \mathbf{Q}(\sqrt{-23})$, $[L : K_0] = 6$, and the relative Galois closure has group S_6. Exact equations and certified branch cycles identify all seven covers. We determine their pointed reductions at 23 and compare the component idempotent with a finite-group invariant. For the distinguished cover, we construct a minimal bidegree-$(23, 4)$ equation, determine the Fano-plane and affine $\mathbf{F}_2^3$-torsor arithmetic of its involution branch fibre, and give a congruence class of M_{23} specializations with controlled ramification. Exact data and verification scripts accompany the results.

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Contents

1. Introduction

Huang, Jackson, Lee, Poonen, Pries, and Zhang [1] constructed a regular M_{23} Galois cover of $\mathbf{P}^1_{\mathbf{Q}}$. Their branch-cycle classes $(2A, 23A, 23B)$ give seven inner Nielsen classes, rather than a rigid triple. Nevertheless, one of the seven classes is fixed by the full arithmetic Galois action. They describe this fixed point as miraculous, noting that they have no conceptual explanation [1, Section 1.4].

We study what distinguishes the source curve of that cover. The answer obtained here is a simple incidence in its canonical embedding. At each of the two totally ramified points, take the osculating plane. The two planes meet in a line. On the distinguished curve this line is a secant; on the other six it misses the curve altogether.

This incidence uses the curve and the unordered pair of marked points. It therefore commutes with Galois conjugation. Once the exceptional incidence has been verified on exactly one of the seven curves, Galois fixedness follows from uniqueness. This supplies a geometric characterization and a computational proof of fixedness.

There is also a local route to existence. We construct a special M_{23}-map in characteristic 23 and follow a lift into canonical coordinates. The essential information is not just the harmonic position of four directions. The two marked sections determine a tame descent character; the effective ramification divisor determines a logarithmic differential; and the two tails must carry the same permutation representation of M_{23}. Together these facts select a local deformation. Exact reconstruction and uniqueness then give the incidence, rather than merely a common zero after reduction.

These arguments have different jobs. The local construction produces an example with the incidence without reading the seven generic models. The global calculation proves that its support in the Hurwitz scheme is a single point. It is this last uniqueness, not local rationality, that gives global Galois fixedness.

1.1. **The global incidence.** Let $\mathcal{H}$ be the finite inner Hurwitz scheme over $\mathbf{Q}$ for the Galois-stable branch datum of [1]: the branch points are $0, \sqrt{-23}, -\sqrt{-23}$, with respective classes $2A, 23A, 23B$. Here *inner* means that generating triples are considered up to simultaneous conjugation by M_{23}. A point of $\mathcal{H}$ represents an M_{23} Galois cover; quotienting by a point stabilizer in its natural degree-23 action gives a genus-4 curve X and a degree-23 map. The latter map is not itself Galois. We call its two totally ramified points A, B.

Over $K_0 = \mathbf{Q}(\sqrt{-23})$, ordering these two branches and normalizing their values to $0, \infty$ identifies $\mathcal{H}_{K_0}$ with the scheme $\mathcal{H}_{\mathrm{in}}$ used in the computations below. The third branch value is then normalized to 1. The seven geometric curves are nonhyperelliptic, so their canonical systems embed them as quadric–cubic complete intersections in $\mathbf{P}^3$.

Theorem 1.1 (Osculating-plane characterization)**.** *On each of the seven canonical curves, the canonical vanishing sequence at A and at B is $(0,1,2,3)$. Let H_A, H_B be their osculating planes, defined by the unique canonical sections vanishing to order at least three at the respective points. Then*

$$\operatorname{length}(X \cap H_A \cap H_B) = \begin{cases} 2, & \textit{on the distinguished cover of } [1], \\ 0, & \textit{on each of the other six covers.} \end{cases}$$

The intersection of length two is reduced and avoids A, B. Consequently the distinguished point of $\mathcal{H}$ is fixed by $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$.

Equivalently, it is the unique cover for which

$$3[A-B] \in X - X \subset \mathrm{Jac}(X), \qquad X - X = \{[C-D] : C, D \in X(\overline{\mathbf{Q}})\}.$$

On this cover the pair (C,D) is unique, and A,B,C,D are distinct. The osculating pencil, after removal of its base divisor, has degree 4 and geometric monodromy S_4; on each other curve its degree is 6.

The proof separates an elementary argument about genus-4 curves from an exact calculation on the seven models. The incidence can also be formed relatively over the Hurwitz scheme: intersect the universal quotient curve with its two relative osculating planes. The resulting finite scheme is supported over one geometric Hurwitz point. This is an actual relative incidence construction in characteristic zero. It is different from the Fano–affine correspondences whose specialization is discussed later.

The existence and descent of the regular M_{23} cover remain due to [1]. Our incidence gives another route to its fixedness, using the reconstructed models to prove that the incidence locus has one-point support. The descent step is standard here: the centralizer of the natural M_{23} action is trivial, so the degree-23 cover has no automorphisms over its base and its descent isomorphisms are unique. We give the details in section 2.

1.2. **Exact arithmetic and reduction.** The global characterization is checked on the complete family, not just on the known rational model.

Theorem 1.2 (Exact arithmetic of the finite inner Hurwitz scheme)**.** *There is a degree-6 field extension L/K_0 and an isomorphism*

$$\mathcal{H}_{\mathrm{in}} \simeq \mathrm{Spec}(K_0 \times L).$$

The relative Galois closure of L/K_0 has group S_6. Over $\mathrm{Spec}(K_0 \times L)$ there is an explicit family of canonical genus-4 curves equipped with degree-23 maps; its seven geometric fibres exhaust the seven Nielsen classes, and each map has monodromy M_{23}.

The exact algebra lets us test the osculating incidence over two number fields. Coprimality over L excludes it simultaneously on all six conjugate curves. Certified branch cycles establish that these curves exhaust the Nielsen class; they also identify the exceptional curve with the one in [1].

Reduction at 23 gives a second description of the same decomposition. The normalized integral Hurwitz model has three closed points, with ramification indices and residue degrees

$$(e,f) = (1,1), \quad (1,2), \quad (2,2).$$

Its reduced closed fibre has algebra $\mathbf{F}_{23} \times \mathbf{F}_{23^2} \times \mathbf{F}_{23^2}$. The complement of the 23-power Frobenius fixed locus lifts to the degree-six component idempotent. The pointed source reductions give an explicit singular-position coordinate for that idempotent.

Theorem 1.3 (Comparison of the component idempotents)**.** *Let e_{osc} be the characteristic function of the nonempty osculating-incidence locus. Let e_{sing} be the idempotent defined in proposition 4.3, and let $\epsilon(\Theta)$ be the finite-group augmentation defined in Exact computation 6.1. In $A_{\mathrm{H}} = K_0 \times L$,*

$$e_{\mathrm{osc}} = (1, 0), \qquad 1 - e_{\mathrm{osc}} = e_{\mathrm{sing}} = \epsilon(\Theta) = (0, 1).$$

This equality is proved by the exact component calculations and branch-cycle identification. It is not a theorem that specialization preserves the osculating incidence: a common zero of two plane sections may appear only in the closed fibre. Nor does it construct a specialization of the finite Fano–affine incidence. The following construction supplies a different input to the incidence argument. Its characteristic-23 equations and finite calculations are given in section 5; no characteristic-zero cover is assumed in constructing its special map.

Theorem 1.4 (A harmonic construction of the incidence)**.** *The two explicit tail covers in equation* (1)*, together with the logarithmic datum in equation* (2)*, define a special M_{23}-map for the branch classes* $(2A, 23A, 23B)$*. After ordering the two wild branches, it has a lift over* $\mathrm{Frac}(W(\overline{\mathbf{F}}_{23}))(\sqrt{-23})$ *whose marked degree-*23 *quotient lies in the canonical neighbourhood equation* (8)*. Its osculating planes meet the generic curve in two reduced points. The lift is identified with the small exact model of theorem 5.9, and hence, by a separate exact comparison, with the distinguished cover of* [1]*.*

This is a computer-assisted existence theorem for a specified special map. The tail Galois groups and the finite jet calculations are explicit inputs; the canonical model and its required jets are conclusions. The theorem neither classifies all stable reductions with this passport nor constructs a Fano–affine specialization correspondence. In particular, it does not make the characteristic-zero component comparison a formal consequence of nearby-cycle duality.

1.3. **Arithmetic on the distinguished cover.** The later sections develop two consequences of having an explicit cover. First, we construct the degree-4 coordinate anticipated in [1, Remark 3.4], giving a primitive bidegree-$(23, 4)$ equation for the same function field. This pencil is $|K_X - A - B|$. It is different from the osculating pencil, whose base divisor avoids A, B.

Second, the fibre above the involution branch point carries a seven-point Fano plane and an affine $\mathbf{F}_2^3$-torsor. We determine the associated $2^4{:}\,\mathrm{PSL}_3(2)$ field tower, its ramification, and its arithmetic flags. The minimal coordinate preserves both residue fields. As a separate application, every integer specialization $t \equiv 2 \pmod{319}$ has Galois group M_{23}; controlling new ramification primes gives an infinite family in which every finite subfamily is linearly disjoint.

The Fano plane lies in one branch fibre of the distinguished cover. It is not the parameter space $\mathcal{H}$. Likewise, the phrase *fibre above the rational branch point* means the fibre above $0 \in \mathbf{P}^1(\mathbf{Q})$, not that its points are rational. We reserve *closed fibre* for a model over a local base.

1.4. **Prior work and organization.** The regular realization, its source equation and branch cycles, the numerical computation of the seven complex covers, and the identification of the Galois-fixed class are due to Huang et al. [1]. Their data [2] are our starting point. The seven-element Nielsen calculation goes back to Häfner [3]. The numerical method is that of Klug–Musty–Schiavone–Voight [4], implemented in the Belyi-map database of Musty–Schiavone–Sijsling–Voight [5].

Our main additions are the exact seven-cover arithmetic, the osculating incidence that characterizes the distinguished point, and the explicit relation with reduction at 23. The curve-theoretic argument uses ordinary canonical linear series and Riemann–Roch; the particular 2 versus 0 incidence is an exact calculation on this family. The local existence argument uses Wewers's lifting theorem for special G-maps [22]. Its additional work is to derive the marked canonical model and compare the two tail actions in one sheet representation, then close the argument by exact algebra and Hensel uniqueness. The auxiliary S_4 map is of interest as a pencil intrinsic to this cover, not as a new realization of S_4.

The point–line Gassmann pair and the abstract incidence geometry of $\mathrm{PSL}_3(2)$ are standard. The contribution here is their arithmetic realization in this branch fibre. The abstract solvability of the split embedding problem with abelian kernel is also known [17]. Huang et al. already obtain an infinite mutually independent family of M_{23}-extensions [1, Corollary 1.4(d)], while general Hilbert–Grunwald methods give progressions with prescribed specialization behavior [15, 16]. Our specialization result specifies one all-members progression and controls new ramification inside it.

Section 2 presents the geometric criterion and its proof. Section 3 supplies the exact equations and their seven-cover identification. Section 4 determines the pointed reductions. Section 5 constructs the harmonic lift and proves its incidence; the local algebra is stated separately so that its hypotheses and its role in the construction remain visible. Section 6 compares the finite invariants and explains the limitations of the separate cohomological approach. The remaining sections treat the minimal equation, branch-fibre arithmetic, specializations, open questions, and certificates.

2. Osculating planes and the distinguished point

We first work with a single smooth nonhyperelliptic genus-4 curve X over an algebraically closed field of characteristic zero. Let $A \neq B$ be points at which the canonical vanishing sequence is $(0,1,2,3)$. In particular,

$$h^0(K_X - 3A) = h^0(K_X - 3B) = 1, \qquad h^0(K_X - 4A) = h^0(K_X - 4B) = 0.$$

Let h_A, h_B be generators of the two one-dimensional spaces, regarded as sections of K_X. Their zero divisors are the sections by the osculating planes H_A, H_B.

In our seven-cover family $h_A(B)$ and $h_B(A)$ are nonzero. Assume these two nonvanishing conditions for the moment. The sections are then independent, and their planes meet in a line ℓ. Put

$$E_{\mathrm{osc}} = \gcd\bigl(\mathrm{div}(h_A), \mathrm{div}(h_B)\bigr), \qquad b_{\mathrm{osc}} = \deg E_{\mathrm{osc}}.$$

On the smooth curve this effective divisor is exactly the scheme-theoretic intersection $X \cap \ell$. The pencil $\langle h_A, h_B \rangle$ has base divisor E_{osc}; after its removal, it defines a map of degree $6 - b_{\mathrm{osc}}$.

2.1. A relation in the Jacobian.

The covering function has divisor $23A - 23B$. Thus $\xi = [A - B] \in \mathrm{Jac}(X)$ has order 23: it is not zero, since a function with divisor $A - B$ would give a degree-one map from a genus-4 curve to $\mathbf{P}^1$.

Lemma 2.1 (Osculation and a difference of points)**.** *Under the preceding hypotheses, suppose $3[A - B] \neq 0$. Then $b_{\mathrm{osc}} \leq 2$, and*

$$b_{\mathrm{osc}} = 2 \iff 3[A - B] = [C - D] \text{ for some } C, D \in X.$$

When these conditions hold, C, D are uniquely determined and

$$\mathrm{div}(h_A) = 3A + D + E_{\mathrm{osc}},$$
$$\mathrm{div}(h_B) = 3B + C + E_{\mathrm{osc}}.$$

In particular A, B, C, D are distinct.

Proof. Write $\operatorname{div}(h_A) = 3A + R_A$ and $\operatorname{div}(h_B) = 3B + R_B$, where $\deg R_A = \deg R_B = 3$. The common divisor avoids A, B, so its degree is at most three. If its degree were three, we would have $R_A = R_B$, giving $3[A - B] = 0$. Hence $b_{\mathrm{osc}} \leq 2$.

If $b_{\mathrm{osc}} = 2$, the residual divisors $R_A - E_{\mathrm{osc}}$ and $R_B - E_{\mathrm{osc}}$ have degree one; call them D, C. Subtracting the canonical divisors gives $[C - D] = 3[A - B]$. Exact order-three contact excludes $D = A$ and $C = B$; nonvanishing at the opposite mark excludes $D = B$ and $C = A$. Finally $C = D$ would contradict $3[A - B] \neq 0$.

Conversely, suppose $3A + D \sim 3B + C$. These are distinct effective divisors of degree four, so their common line bundle $\mathcal{L}$ has $h^0(\mathcal{L}) \geq 2$. Riemann–Roch gives

$$h^0(K_X \otimes \mathcal{L}^{-1}) = h^0(\mathcal{L}) - 1 \geq 1.$$

Choose an effective divisor E' of degree two in $|K_X \otimes \mathcal{L}^{-1}|$. Then $3A + D + E'$ and $3B + C + E'$ are canonical divisors. Uniqueness of h_A, h_B forces E' to be common to their zero divisors. Thus $b_{\mathrm{osc}} \geq 2$, and the preceding bound gives $E' = E_{\mathrm{osc}}$. Subtraction now also proves uniqueness of C, D. □

There is an equivalent test using a line bundle of degree six. For a nontrivial degree-zero class η, Riemann–Roch gives $h^0(K_X + \eta) = 3$, and a point C is a base point of this linear system precisely when

$$h^0(K_X + \eta - C) = 2 + h^0(C - \eta) = 3.$$

This is equivalent to $\eta = [C - D]$ for a point D. Taking $\eta = 3\xi$ gives the criterion in lemma 2.1. The osculating-plane test is computationally smaller: it needs only canonical jets through order three and a polynomial gcd on a line.

2.2. **The exact incidence calculation.** The input consists of the canonical quadric q, the cubic c, and the two marked points on each coefficient component of the exact Hurwitz algebra. The stored data call these marks b, c; here $A = b$ and $B = c$. Their smoothness and their identification with all seven covers are established in section 3. The following calculation uses no reduction modulo 23.

Exact computation 2.2 (Osculating incidence on all seven models)**.** Over K_0 on the degree-one component and over L on the sextic component, exact canonical jets at both marks have ranks 3 through order two and 4 through order three. Their one-dimensional kernels through order two define h_A, h_B, and $h_A(B)h_B(A) \neq 0$. Restricting the canonical equations to $\ell = H_A \cap H_B$ gives binary forms q_ℓ, c_ℓ with

coefficient component	$\deg \gcd(q_\ell, c_\ell)$	deg osculating map	$3[A - B] \in X - X$
K_0	2	4	yes
L	0	6	no.

In the first row q_ℓ is squarefree and divides c_ℓ. In the second row the two binary forms are coprime.

Exact certificate. Choose an affine chart containing a marked point and a local parameter t. An invertible 2×2 Jacobian minor of (q, c) solves the other two coordinates successively modulo t^4. The coefficients of $1, t, t^2$ in the four canonical coordinates give a 3×4 matrix. Its kernel gives the osculating form. Appending the coefficient of t^3 gives an invertible matrix, proving both the vanishing sequence and exact order-three contact. The two resulting linear forms give a basis for their common two-dimensional kernel in the coordinate vector space, hence a homogeneous parametrization of ℓ.

Substitute that parametrization into q, c and compute their homogeneous gcd over the coefficient field. In both cases q_ℓ is a nonzero quadratic, so the common zero scheme on $\mathbf{P}^1$ has length equal to the gcd degree. This computation includes points at infinity. The nonzero discriminant in the degree-one case proves reducedness. Coprimality over L is preserved under every embedding of L, so it excludes incidence on all six geometric conjugates.

For a compact record of the first row, put $s = \sqrt{-23}$. In the stored canonical coordinates,

$$h_A = x_3,$$
$$h_B = x_0 + \frac{387s - 51}{1508}x_1 + \frac{4851s - 9547}{19604}x_2 + \frac{30411308s - 56608155}{53582633}x_3.$$

On their intersection line the common zero scheme is given, up to a nonzero scalar, by

$$x_1^2 + \frac{387s + 1457}{1508}x_1x_2 - \frac{217389s + 189591}{568516}x_2^2 = 0.$$

Its discriminant is nonzero. The certificate `verification/verify_hurwitz_osculating.py` recomputes both rows, checks the residual points of lemma 2.1, and repeats the incidence calculation after an invertible projective change of coordinates. The input hashes are recorded in section 11. □

2.3. **The relative incidence and Galois fixedness.** The preceding construction is algebraic in families. For the reconstructed universal quotient curve $f : \mathcal{X} \to \mathcal{H}_{\mathrm{in}}$, let $\mathcal{V} = f_*\omega_{\mathcal{X}/\mathcal{H}_{\mathrm{in}}}$. It has rank four. Restriction to the length-three subschemes $3A$ and $3B$ gives maps

$$\mathcal{V} \longrightarrow f_*(\omega|_{3A}), \qquad \mathcal{V} \longrightarrow f_*(\omega|_{3B}).$$

Both targets have rank three. The ranks in Exact computation 2.2 show that the kernels are line bundles and that their formation commutes with field extension. Their evaluation on $\mathcal{X}$ defines the two relative osculating plane sections.

Proposition 2.3 (Relative osculating incidence)**.** *Let $\mathcal{Z} \subset \mathcal{X}$ be the common zero scheme of those two sections. Then $\mathcal{Z} \to \mathcal{H}_{\mathrm{in}}$ is finite. It is reduced of degree two over the degree-one component and empty over the sextic component. Its scheme-theoretic image is the degree-one open-and-closed subscheme.*

The construction is unchanged by exchange of A, B and descends to the original rational branch datum. Its image over $\mathcal{H}$ is a single $\mathbf{Q}$-rational point.

Proof. The construction commutes with isomorphisms of marked curves and with field extension, since it uses the canonical bundle, restriction to $3A, 3B$, and their kernel subspaces. The map $\mathcal{Z} \to \mathcal{H}_{\mathrm{in}}$ is proper with finite geometric fibres, hence finite. Exact computation 2.2 gives its fibre lengths and reducedness. Because the base is a product of fields, its scheme-theoretic image is exactly the component on which the fibre is nonempty.

The Galois action on the original rational branch datum preserves the unordered pair of totally ramified points. It therefore preserves $\mathcal{Z}$ and its image: swapping the marks merely exchanges the two defining ideals. The corresponding construction over $\mathcal{H}$ can be made after the extension $K_0/\mathbf{Q}$ and descended, or formed directly from the universal quotient curve using the unordered pair. Its image has exactly one geometric point. A reduced finite scheme in characteristic zero with one geometric point is $\operatorname{Spec}\mathbf{Q}$. □

For completeness, this fixed point also descends as a cover. A deck automorphism of a connected degree-23 cover acts on a geometric unramified fibre by a permutation centralizing its monodromy. The natural M_{23} action is 2-transitive and its centralizer in S_{23} is trivial: a point stabilizer fixes no other letter, so a commuting permutation fixes one letter and then every letter. Thus isomorphisms of the cover with its Galois conjugates are unique. They satisfy the cocycle condition and give effective descent to $\mathbf{Q}$.

Its arithmetic monodromy normalizes the geometric M_{23}. The identity $N_{S_{23}}(M_{23}) = M_{23}$, also checked in proposition 8.18, implies that arithmetic and geometric monodromy agree. Hence its

Galois closure is regular over $\mathbf{Q}$. This recovers the known conclusion of [1] from the new incidence computation; it does not claim a new regular realization.

2.4. **A second degree-four pencil.** On the exceptional curve write $E = E_{\mathrm{osc}}$. The ratio h_B/h_A has divisor

$$3B + C - 3A - D.$$

It gives a degree-4 map with ramification index three at both A, B. The pencil is intrinsic to $(X, \{A, B\})$, so on the descended $\mathbf{Q}$-curve it is a Galois-stable two-dimensional subspace of $H^0(X, K_X)$. Linear descent gives a two-dimensional $\mathbf{Q}$-space and hence a map to $\mathbf{P}^1_{\mathbf{Q}}$.

Proposition 2.4 (Monodromy of the osculating pencil)**.** *The geometric monodromy of this degree-4 map is S_4. Its Galois closure over its rational target is therefore regular. The pencil is distinct from $|K_X - A - B|$.*

Proof. In the stored coordinates use the affine chart $x_3 = 1$ and set $w = h_B/h_A$. The equation $h_B = w$ eliminates x_0. Eliminating x_1 between the resulting quadric and cubic gives a primitive quartic in x_2 over $K_0[w]$. A linear subresultant recovers x_1, with coefficient invertible modulo the quartic; substitution annihilates both original equations. Thus the quartic represents the actual function field of X over $K_0(w)$, not an extra resultant component.

The discriminant has squarefree factors whose (degree, multiplicity) pairs are

$$(10, 1), \qquad (5, 2).$$

In particular it is not a square even after extending the constants to $\overline{\mathbf{Q}}$. These elimination, subresultant, and discriminant identities are recomputed by the option `--quartic-monodromy` of the osculating certificate.

The geometric monodromy is transitive of degree four and contains a 3-cycle from the ramification at A. It is therefore A_4 or S_4. The nonsquare discriminant excludes A_4. After descent to $\mathbf{Q}$ the arithmetic group cannot be larger than S_4, so its equality with the geometric group proves regularity.

Finally E and $A + B$ are distinct effective degree-two divisors: the former avoids both marked points. If they were linearly equivalent, their moving linear system would give a degree at most two map to $\mathbf{P}^1$, contrary to nonhyperellipticity. Thus $K_X - E$ and $K_X - A - B$ are distinct, as are the two pencils. $\square$

Proof of theorem 1.1. The jet and incidence assertions are Exact computation 2.2; their application to all seven covers uses corollary 3.7. Lemma 2.1 proves the Jacobian criterion and uniqueness of the residual points. Proposition 2.3 proves global Galois fixedness. The degree of the pencil is $6 - \deg E_{\mathrm{osc}}$, and proposition 2.4 gives its exceptional monodromy. $\square$

The remaining existence issue is now concrete. We have certified that the relative incidence has nonempty fibre on exactly one cover. We have not shown, without these equations, that the branch datum forces this to happen. An intersection-theoretic or specialization argument establishing that fact would strengthen the characterization into a more structural explanation.

3. Exact reconstruction of the seven covers

3.1. **Coordinate algebra and canonical models.** This section supplies the complete set of exact models used in the osculating calculation. Numerical reconstruction discovers the coefficients; exact function-field identities and certified branch cycles identify the resulting covers. The normalized canonical quadric has the normalization-invariant coefficient

$$J = \frac{q_{13}}{A^2},$$

where A is the coefficient used to impose the fixed normalization in the accompanying data. Computing J at the seven analytic classes and reconstructing over K_0 gives a finite étale K_0-algebra

$$A_{\mathrm{H}} = K_0 \times L, \qquad [L : K_0] = 6.$$

Let $\mathcal{H}_J = \operatorname{Spec} A_{\mathrm{H}}$. Its degree-one component carries the reconstructed map belonging to the distinguished class, and the six geometric points of its connected degree-six component carry the other six reconstructed maps. The branch-cycle computation below identifies $\mathcal{H}_J$ with the inner Hurwitz scheme $\mathcal{H}_{\mathrm{in}}$.

Exact computation 3.1 (Coordinate algebra and canonical curves)**.** The linear factor of the polynomial of J is

$$J - \frac{148227 - 34830\sqrt{-23}}{142129},$$

and its complementary factor is irreducible of degree 6 over K_0. An absolute defining polynomial for L is

$$\begin{aligned} h(Z) = Z^{12} - 6Z^{11} + 20Z^{10} - 32Z^9 + 44Z^8 - 22Z^7 + 6Z^6 \\ - 22Z^5 + 44Z^4 - 32Z^3 + 20Z^2 - 6Z + 1. \end{aligned}$$

Over A_{H}, all ten coefficients of the canonical quadric, all twenty coefficients of a Petri cubic in the fixed coefficient normalization, and the two marked order-23 points reconstruct exactly. On each component the resulting quadric–cubic complete intersection is smooth.

The reciprocal form of h makes the Galois closure unexpectedly small to describe. Put

$$g(Y) = Y^6 - 6Y^5 + 14Y^4 - 2Y^3 - 27Y^2 + 44Y - 44.$$

Proposition 3.2 (Galois closure of the sextic component)**.** *Let $E = \mathbf{Q}[Y]/(g)$ and let $\widetilde{E}$ be its Galois closure. Then*

$$L = EK_0, \qquad \mathrm{Gal}(\widetilde{E}/\mathbf{Q}) \cong S_6, \qquad \mathrm{Gal}(\widetilde{E}K_0/K_0) \cong S_6,$$

and

$$\mathrm{Gal}(\widetilde{E}K_0/\mathbf{Q}) \cong S_6 \times C_2.$$

The relative Galois group acts as the full symmetric group on the six remaining maps. In the embedding order of Exact computation 3.6, its letters have Nielsen labels

$$(7, 4, 1, 5, 3, 2).$$

Moreover

$$\mathrm{disc}(E) = 2^4 \cdot 11 \cdot 23^4, \qquad \mathfrak{d}_{L/K_0} = (2^4 \cdot 11 \cdot 23).$$

Proof. Let a be the image of Z in $\mathbf{Q}[Z]/(h)$. The reciprocal-polynomial identity is

$$h(Z) = Z^6 g(Z + Z^{-1}).$$

The exact relative-to-absolute field map sends $\sqrt{-23}$ into $\mathbf{Q}(a)$, and the element $a + a^{-1} + \sqrt{-23}$ has degree 12 over $\mathbf{Q}$. Consequently $L = EK_0$. Maximal-order calculations give the displayed discriminants.

The rational primes $3, 139, 2671$ split in K_0, do not divide $\mathrm{disc}(g)$, and give respective factor-degree patterns

$$(6), \qquad (1, 5), \qquad (1, 1, 1, 1, 2)$$

for g modulo those primes. Thus the relative Galois group contains a 6-cycle, a 5-cycle, and a transposition. The first factorization also proves irreducibility. Of the sixteen transitive subgroups of S_6, only $6T16 = S_6$ contains all three cycle types.

Finally, the unique quadratic subfield of an S_6 splitting field is its discriminant field. Since

$$\mathrm{disc}(g) = 2^{22} \cdot 11 \cdot 23^4,$$

that field is $\mathbf{Q}(\sqrt{11})$, rather than $K_0 = \mathbf{Q}(\sqrt{-23})$. Therefore $\widetilde{E}$ and K_0 are linearly disjoint, giving the two asserted Galois groups. The labeling by Nielsen classes is obtained from the certified branch-cycle continuation. ☐

3.2. **Reconstruction and identification of the seven maps.** We now pass from the reconstructed coordinate algebra to the maps themselves. The analytic calculation supplies a uniform error bound; exact canonical-ring linear algebra then reconstructs the ratios, and certified branch cycles identify their moduli points.

Exact computation 3.3 (Uniform truncation-error certificate)**.** The 47 chart centres cover both half-triangles with pseudohyperbolic radius less than 0.471: an Arb calculation [10] on geodesically convex mesh cells in the Klein disk model gives the upper bound 0.4708683715. At Cauchy radius $\rho = 0.72$, the modes $0, \dots, 60$, augmented by four normalization rows, have certified minimum singular value at least $6.60 \cdot 10^{-4}$ in all seven classes. Let A be the augmented low-mode matrix, h the norm bound for the high-mode operator, and ℓ the bound for the low-mode transition operator. Parseval gives $h \leq 2.35 \cdot 10^{-11}$ and $\ell \leq 1.38 \cdot 10^3$, so the explicitly defined Schur-complement margin

$$m = \sigma_{\min}(A) - \frac{h\ell}{1-h}$$

is positive in every class.

A bootstrap using Cauchy estimates at outer radius 0.99 bounds the exact normalized forms. The final $N = 700$, $Q = 1280$ Acb calculation then evaluates all 1280 discrete Fourier transform (DFT) output modes, not only those through the polynomial cutoff. Combining its residual balls with the certified inverse and branch-jet conditioning gives at least 75 certified decimal digits in each of the first twenty branch-normalized Taylor rows used to recover the canonical quadric and Petri cubic.

The exact equations are too large to reproduce usefully in print. They are stored in power or integral bases in `hurwitz_canonical_models_candidate.json` and `hurwitz_marked_points_candidate.json`. Their importance here is that they allow the covering function to be recovered by linear algebra internal to the canonical ring, with no search for a low-degree plane ansatz.

Proposition 3.4 (Exact degree-23 ratios)**.** *Let $C \to \operatorname{Spec} A_{\mathrm{H}}$ be the reconstructed family of canonical complete intersections, and let $b, c : \operatorname{Spec} A_{\mathrm{H}} \to C$ be the two sections marking the points with order-23 inertia. There are explicit quintic canonical sections N, D such that*

$$u = \frac{N}{D}, \qquad \mathrm{div}(u) = 23b - 23c.$$

In particular u has degree 23 on every geometric fibre. Restriction to the degree-6 factor of A_{H} and its six K_0-embeddings gives six explicit conjugate maps.

Proof. The degree-5 piece of the canonical ring has dimension 27. The first 23 local coefficients at either marked point have rank 23, and hence

$$B = H^0(5K_C - 23b), \qquad C' = H^0(5K_C - 23c)$$

both have dimension 4. The degree-10 piece has dimension 57. Multiplication there turns the identities

$$B_i(MC')_j - B_j(MC')_i = 0 \qquad (0 \leq i < j \leq 3)$$

into a linear system with 16 unknowns. Its rank is 15; its kernel is generated by an invertible matrix M. The deterministic first row gives $N = B_0$ and $D = (MC')_0$. Exact local expansions give

$$\operatorname{ord}_b(N) = 23, \quad D(b) \neq 0, \qquad \operatorname{ord}_c(D) = 23, \quad N(c) \neq 0.$$

Finally, saturating the ideal generated by the four sections of B away from b gives the unit ideal, and the same calculation for MC' away from c again gives the unit ideal. Thus the two linear systems have no base points away from b and c. Together with the displayed local orders, the fixed divisors of these complete linear systems are exactly $23b$ and $23c$. Write $G_i = (MC')_i$. The multiplication identities imply $B_i = uG_i$ for one rational function $u = N/D$. Taking the minimum of the orders over i at every point gives

$$\operatorname{div}(u) = 23b - 23c.$$

The chosen sections themselves do have a common effective divisor:

$$\operatorname{div}(N) = 23b + E, \qquad \operatorname{div}(D) = 23c + E, \qquad \deg E = 7.$$

Indeed, a quintic canonical section has degree $5(2g-2) = 30$, so the common divisor is necessary. Cancelling it gives the asserted degree-23 map. □

It remains to put the third branch point at 1. The raw value is $\lambda = u(a) \in L$, where a is any point above the order-2 branch point. Its coordinates in the chosen power basis have a large common denominator, so ordinary coordinatewise recognition is ineffective.

Exact computation 3.5 (Third branch normalization)**.** At 71 rational primes splitting completely in L, the unique nonzero linear factor of multiplicity 8 in the reduced branch discriminant is computed in all twelve residue embeddings and interpolated into the power basis. Their CRT modulus has 404 decimal digits. A 13-dimensional LLL calculation [9] gives a 1129-bit common-denominator vector; the next lattice vectors have at least 1246 bits. This value agrees with the independent 1024-bit Arb computation to $1.7 \cdot 10^{-101}$ in integral-basis coordinates. It also agrees exactly at 24 further completely split primes that were withheld from CRT and LLL.

For a characteristic-zero check, project the canonical complete intersection birationally to a plane sextic. After cancellation, N and D become plane forms of degree 7. The resultants of the plane sextic with a generic fibre and with $N - \lambda D$ both have degree 42. Their gcds with their x-derivatives have degrees 6 and 14, respectively; the common factor has degree 6, and the additional degree-8 quotient is squarefree. The only other possible source of such a repeated projected root is two distinct plane points with the same x-coordinate. Magma excludes this after good reduction at the completely split prime 863153: in residue embeddings $2, 3, 4, 5, 9, 12$, every irreducible factor of the degree-8 quotient gives a degree-1 gcd in the remaining coordinate. Since the resultant, gcd, and squarefree-factor degrees are preserved, a characteristic-zero common divisor of degree at least 2 would remain one after reduction. Thus the third fibre has eight distinct double points and seven simple points. Independently, the projective critical ideal has length $15 = 7 + 8$ in all twelve residue embeddings.

With this value, the six maps are normalized by

$$\beta = \frac{N}{\lambda D};$$

their branch values are $0, 1, \infty$, with passport

$$(23), \qquad (2^8 1^7), \qquad (23).$$

The exact field element λ, all reconstruction and holdout residues, and the sparse sections $N, D, \lambda D$ are stored in `hurwitz_degree23_branch_candidate.json` and `hurwitz_degree23_maps_candidate.json`. The degree and the two totally ramified fibres in proposition 3.4 are characteristic-zero exact statements, as is the third-fibre multiplicity in Exact computation 3.5. The uniform Taylor-tail estimate is supplied by Exact computation 3.3. The filenames retain the word *candidate* to record that LLL and bounded-denominator reconstruction were the discovery route for the displayed coefficients; the following independent cover identification removes that reconstruction assumption from the final Hurwitz-algebra statement.

Exact computation 3.6 (Exact eliminant and certified branch cycles)**.** In the affine canonical chart $X_0 = 1$, eliminate $X_3 = z$ from the quadric and Petri cubic. If their linear subresultant is $L_1(x, y)z + L_0(x, y)$, then every serialized quintic is linear in z, so a section $S_0 + zS_1$ becomes the degree-7 plane form $S_0L_1 - S_1L_0$. Eliminating y from the plane sextic and $N - t\lambda D$ gives a raw resultant $R_t(x)$ of degree 42. The exact gcd of R_2 and R_3 has degree 19 and divides every R_t; after division, the values at $t = 2, \ldots, 8$ interpolate

$$P(t, x) \in L[t, x], \qquad \deg_t P = 6, \quad \deg_x P = 23.$$

The unused value $t = 9$ agrees exactly. At $t = 0, 1, \infty$, the gcds with the x-derivative have degrees $22, 8, 22$, and the degree-8 gcd is squarefree.

For each of the six embeddings of L over the chosen embedding of K_0, Arb isolates the 23 roots over $t_* = 1/2 + 2i$ and continues them around rational polygonal loops about $0, 1, \infty$. On every straight path segment, each sheet is enclosed in a disk of radius R with fixed inverse slope Y. Writing

$$\eta = \sup |YP(t, c)|, \qquad q = \sup |1 - YP_x(t, x)|,$$

the segment is accepted only when $q < 1$ and $\eta + qR < R$, all 23 disks are pairwise disjoint, and the endpoint root balls lie in their unique disks. Otherwise the segment is bisected. Thus the continuation is certified by a uniform contraction, not by nearest-neighbour matching. The map on the degree-one component (Nielsen class 6) is checked independently from its integral minimal-degree equation by the same method.

The full run contains 150145 accepted tubes and has maximum dyadic depth 18. In the sextic-embedding order the Nielsen IDs are

$$(7, 4, 1, 5, 3, 2).$$

Together with the independently checked degree-one map, whose Nielsen ID is 6, the seven triples have cycle shapes

$$(23), \qquad (1^7 2^8), \qquad (23),$$

their products are the identity, and every pair generates a permutation group of order 10200960. After conjugating the order-23 generator to the fixed y, centralizer translation matches the seven involutions with Nielsen representatives $1, \ldots, 7$, exactly once each. The independent normalization of the degree-one map orders the two order-23 endpoints oppositely, so its comparison uses the inverse of its loop about 0.

Corollary 3.7 (Identification of the finite Hurwitz scheme)**.** *The finite étale scheme* $\mathcal{H}_J = \mathrm{Spec}(K_0 \times L)$ *is isomorphic to* $\mathcal{H}_{\mathrm{in}}$*. Thus* $A_{\mathrm{H}} = K_0 \times L$ *is the coordinate algebra of the finite inner Hurwitz scheme of degree* 7 *and type* $(2A, 23A, 23B)$*. In particular, all seven reconstructed maps have monodromy* M_{23}*.*

Proof. The group-side Nielsen calculation gives exactly seven inner classes. Exact computation 3.6 produces one exact cover in each of them. They are therefore distinct and exhaustive. The induced map between the two reduced finite étale K_0-schemes is a bijection on geometric points, hence an isomorphism. This identifies the coordinate algebra without an a priori height bound for the preceding LLL discovery. □

Proof of theorem 1.2. The coordinate algebra and smooth canonical curves are given by Exact computation 3.1. The exact maps are constructed in proposition 3.4 and Exact computation 3.5; their certified branch cycles give the identification with $\mathcal{H}_{\mathrm{in}}$ in Exact computation 3.6 and corollary 3.7. Finally, proposition 3.2 proves the S_6 assertion. □

4. Reduction at 23

The osculating incidence is a global construction in characteristic zero. Here we describe how the same distinguished component appears in the pointed reduction at 23 and in finite-group calculations. The exact algebra and branch-cycle certificates will compare their values. A direct specialization between the underlying geometric constructions is not assumed.

4.1. **Local structure at 23.** The prime above 23 gives a particularly small local realization of this full symmetric action. Let $\mathfrak{q} = (\sqrt{-23})$ be the unique prime of K_0 above 23.

Proposition 4.1 (Local Hurwitz algebra at 23)**.** *The normalization of* $\operatorname{Spec} \mathcal{O}_{K_0,\mathfrak{q}}$ *in* $A_{\mathrm{H}} = K_0 \times L$ *has three closed points. Their ramification indices and residue degrees over* $K_{0,\mathfrak{q}}$ *are*

$$(e, f) = (1, 1), \qquad (1, 2), \qquad (2, 2),$$

where the first point lies on the degree-one component and the other two lie on the sextic component. In the S_6*-closure, a decomposition group at* $\mathfrak{q}$ *is*

$$D_{\mathfrak{q}} \cong V_4.$$

Its natural action on the six remaining points has orbits of sizes 2 *and* 4*; inertia has cycle type* $1^2 2^2$*, and either lift of residue Frobenius has cycle type* 2^3*.*

Proof. Modulo 23 the trace sextic satisfies

$$g(Y) = (Y - 5)^2 (Y + 1)^4.$$

Over $\mathbf{Q}_{23}$ its two factors have degrees 2 and 4. Their discriminant valuations are 1 and 3, with residual units 9 and 7; their discriminant square classes are therefore 23 and -23. The quartic factor is totally tamely ramified. Its local Galois closure is dihedral of order 8: tame inertia is cyclic of order 4, while Frobenius acts by inversion because $23 \equiv -1 \pmod 4$. Its discriminant field is $\mathbf{Q}_{23}(\sqrt{-23})$. Restriction to $K_{0,\mathfrak{q}}$ gives the even Klein four subgroup. The quadratic factor field is $\mathbf{Q}_{23}(\sqrt{23})$; its compositum with $K_{0,\mathfrak{q}}$ is the same unramified quadratic field already contained in the dihedral closure. This gives the asserted group action and prime data.

The first residual equations make the common residue field explicit. If $s = \sqrt{-23}$, then

$$U = \frac{Y - 5}{s}, \quad \overline{U}^2 = 15, \qquad V = \frac{(Y + 1)^2}{s}, \quad \overline{V}^2 = 5.$$

Both equations define $\mathbf{F}_{23^2}$, since $15/5 = 3 = 7^2$ in $\mathbf{F}_{23}$. □

A complex Nielsen labeling of the local $2 + 4$ partition is not canonical: it requires choosing a prime of the S_6-closure above $\mathfrak{q}$, and changing that prime conjugates $D_{\mathfrak{q}}$ inside S_6.

We use the standard ADE nomenclature and the Milnor, Tjurina, and delta invariants as in [11].

Exact computation 4.2 (Pointed closed fibres at 23)**.** The exact canonical equations, marked points, and degree-23 map sections are integral at each of the three closed points of the normalized local Hurwitz scheme. The resulting closed fibres are geometrically integral quadric–cubic complete intersections, their canonical quadrics remain smooth, and the marked zero b and pole c remain distinct smooth points. Their singularities are

closed point	residue field	singularities
degree-one point (class 6)	$\mathbf{F}_{23}$	E_8
point with $(e, f) = (1, 2)$	$\mathbf{F}_{23^2}$	E_8
point with $(e, f) = (2, 2)$	$\mathbf{F}_{23^2}$	$A_2 + A_6$.

The Milnor and Tjurina numbers are respectively 8, and $2 + 6$. In each row the total delta invariant is 4, so the geometric normalization is $\mathbf{P}^1$.

For the sextic component let N, D denote the reduced numerator and normalized denominator. For the degree-one component use the stored numerator and denominator, whose third branch value has not yet been scaled to one. Write $v = N/D$ for these stored ratios. Their common base scheme has length 7, avoids the singular points, and

$$\operatorname{ord}_b(N) = \operatorname{ord}_c(D) = 23, \qquad \operatorname{ord}_b(D) = \operatorname{ord}_c(N) = 0.$$

Consequently, on the normalization,

$$\operatorname{div}(N/D) = 23b - 23c.$$

There is a unique coordinate t on the normalization with $b = 0$, $c = \infty$, and

$$v = t^{23}$$

at all three closed points. This t is the coordinate for the stored ratio v, not yet for a uniformly normalized Belyi map on the degree-one component.

Proof. Exact reduction in the two residue fields gives the displayed singular loci and no singularity at infinity. Eliminating the smooth-quadric coordinate gives an irreducible plane sextic. The exact local plane equations have triple tangent, Milnor–Tjurina pair $(8, 8)$, and fifth-order tangent restriction in the first two cases, identifying E_8 in characteristic 23. At the third point the two completed local equations have Milnor–Tjurina pairs $(2, 2)$ and $(6, 6)$ and critical-value orders 3 and 7, identifying A_2 and A_6. Their delta invariants exhaust the arithmetic genus 4. The common base ideal has constant Hilbert polynomial 7, and exact local series at b, c give the four displayed orders. Since the residue fields are perfect, the remaining scalar multiplying t^{23} can be absorbed into t. The resulting coordinate is unique, since $a^{23} = 1$ in characteristic 23 implies $a = 1$. □

These pointed closed fibres contain more arithmetic information than their ADE types alone. If P is one of the three closed points, let τ_P be the unique singularity in the E_8 cases and the uniquely characterized A_6 singularity in the third case. Both are unibranch. We may therefore evaluate the stored-ratio normalization coordinate at the point above τ_P. We distinguish it from the uniformly branch-normalized coordinate in proposition 4.4 below.

Proposition 4.3 (Idempotent defined by the normalized singularity coordinate)**.** *Put $u(P) = t(\tau_P)$. On the reduced closed fibre of the normalized local Hurwitz scheme, u is a primitive element of its coordinate algebra, with minimal polynomial*

$$R_{23}(u) = (u-16)(u^2+1)(u^2+u+1) = 0.$$

The linear factor is the degree-one E_8 point, the factor u^2+1 is the unramified quadratic E_8 point, and u^2+u+1 is the ramified A_2+A_6 point. In

$$B = \mathbf{F}_{23}[u]/(R_{23})$$

the polynomial

$$e_{\text{sing}}(u) = 2u^4 + 2u^3 + 4u^2 + 2u + 3$$

is idempotent and has values $(0,1,1)$ on these three factors. Its unique lift to the completed local Hurwitz algebra is the sextic component idempotent $e_6 = (0,1)$. In particular,

$$e_{\text{sing}}(P) = f(P) - 1 \pmod 2.$$

Proof. Exact evaluation on the normalizations gives

closed point	$u(P)$	minimal polynomial over $\mathbf{F}_{23}$
degree one, E_8	16	$u-16$
$E_8,\ (e,f)=(1,2)$	$\sqrt{-1}$	u^2+1
$A_6,\ (e,f)=(2,2)$	r	$r^2+r+1.$

In the last row the A_2 point has parameter $t=1$, while Frobenius exchanges r and r^2. The constructions of t and τ_P commute with residue-field automorphisms, so the displayed values define an element of the coordinate algebra of the reduced closed fibre. The three factors are pairwise coprime, and their degrees sum to the five geometric points of that fibre, proving the primitive-coordinate assertion.

If $A = (u^2+1)(u^2+u+1)$, then $A(16) = 11$ and $11^{-1} = 21$ in $\mathbf{F}_{23}$. Hence

$$e_{\text{sing}} = 1 - 21A = 1 + 2A,$$

which gives both its displayed formula and its three values. Idempotents lift uniquely through nilpotent thickenings and complete local rings. The two quadratic factors are precisely the two local factors of L, so the lift is e_6. □

Proposition 4.4 (Uniform branch normalization and the harmonic point)**.** *On the degree-one canonical model, the third branch value λ of the stored ratio $v = N/D$ is a unit at 23, with residue 7. On this component put $\beta = v/\lambda$; on the other two components put $\beta = v$, since their stored maps are already normalized. Let t_{n} be the normalization coordinate with $\overline{\beta} = t_{\mathrm{n}}^{23}$. The distinguished singular positions in this coordinate have polynomial*

$$(t_{\mathrm{n}}+1)(t_{\mathrm{n}}^2+1)(t_{\mathrm{n}}^2+t_{\mathrm{n}}+1).$$

Thus the degree-one point is characterized by $t_{\mathrm{n}} = -1$. Relative to the three normalization points $0, \infty, 1$, its singular point is their harmonic conjugate: the ordered cross-ratio $(0,\infty;1,-1)$ is -1.

Proof. On the affine canonical model, form the critical ideal of N/D by adjoining the determinant of the two equation-gradient rows and the row $D\,dN - N\,dD$. Saturate away from $ND = 0$. In this nonempty finite critical algebra, exact reduction gives $N = \lambda D$ for a scalar $\lambda \in K_0$. Since the third branch value is the only branch value other than zero and infinity, this determines it. The certificate `verification/verify_hurwitz_degree_one_branch_normalization.py` recomputes this scalar and checks that its valuation is zero and its residue is 7.

Inverse Frobenius in $\mathbf{F}_{23}$ fixes 7, so $t_{\mathrm{n}} = t/7$ on the degree-one reduction. Its old singular position 16 becomes $16/7 = -1$. The other two stored maps require no scaling, giving the displayed polynomial. The marked points reduce to $0, \infty$; the unique point of the normalization over the third branch value has $t_{\mathrm{n}} = 1$. This proves the cross-ratio interpretation. $\square$

The polynomial in proposition 4.3 is retained to agree with the stored-ratio certificate. The normalized polynomial above describes the same residue algebra in another primitive coordinate. Neither the component idempotent nor the residue degrees change. This local harmonic condition and the global osculating incidence select the same component by exact calculation. Harmonic position alone does not specify a mixed-characteristic deformation. In section 5 we instead construct a special map with this position, derive its canonical deformation, and prove incidence on its lift.

Thus the characteristic-23 models themselves construct the morphism to the constant two-point scheme that cuts out the degree-six component; no complex Nielsen labels or continued branch cycles enter proposition 4.3. This construction alone does not yet identify the two expressions

$$\epsilon(\Theta) = e_{\mathrm{sing}}.$$

The exact branch-cycle certificate proves the equality indirectly because both sides are e_6; this is the proof used in theorem 1.3. Section 6.3 then asks whether the equality can also be obtained from a single specialization argument. Resolving only the coarse ADE source is insufficient: the degree-one and unramified quadratic reductions both have geometric type E_8. A successful geometric proof must retain the arithmetic singular position together with compatible incidence and gluing data.

5. Constructing the harmonic lift

We now start in characteristic 23, rather than reduce a known generic equation. The purpose is to produce a cover with the osculating incidence. The construction has three stages: lift a special M_{23}-map, derive its marked canonical neighbourhood, and identify the lift by exact local uniqueness. The first two stages are developed here; the last is the local reconstruction theorem below.

Throughout this section put

$$k = \overline{\mathbf{F}}_{23}, \qquad K^{\mathrm{ur}} = \mathrm{Frac}(W(k)), \qquad K = K^{\mathrm{ur}}(\pi), \qquad \pi^2 = -23,$$

and let $\mathcal{O}$ be the valuation ring of K, with $v(\pi) = 1$. Write $G = M_{23}$, let $M = M_{22}$ be a point stabilizer, and choose $P \simeq C_{23}$ and $D = N_G(P) = P{:}H$, where $H = M \cap D \simeq C_{11}$. The letter D in this section denotes this normalizer, not a residual point in the divisor relation of theorem 1.1.

5.1. The special map and its lift.

A *tail* is a component meeting the rest of a semistable target in one point. A primitive tail contains a specialization of an original tame branch point; a new tail contains none. The associated covers below are separable degree-23 maps; their normal closures, not the quotient maps themselves, are G-covers.

Exact computation 5.1 (Two characteristic-23 tail covers)**.** Over $\mathbf{F}_{23}$ consider

$$\begin{aligned} D_7 :&\quad T = X^{23} + 7X^9 + 18X^2, \\ D_{15} :&\quad w^3 = 19(X^5 + 17), \qquad T = (X^6 + 2X)w. \end{aligned} \tag{1}$$

Their normal closures have geometric Galois group M_{23}. At infinity the inertia groups are 23:11, with lower jumps 7 and 15, respectively. The first source has genus zero and finite branch fibre $2^8 1^7$; the second has genus four and is étale over the affine target.

Here the group assertion is a function-field computation, recorded separately for the two equations in section 11. The arithmetic groups over $\mathbf{F}_{23}(T)$ are M_{23}. Their geometric subgroups are normal and contain nontrivial inertia, so simplicity gives the same geometric groups. The remaining assertions can be checked directly. For the first map,

$$T = (X^8 + 17X)^2(X^7 + 12), \qquad T' = 17(X^8 + 17X),$$

with squarefree, coprime factors. Its different at infinity has degree $44 - 8 = 36$. For the second, put $H_1 = X^6 + 2X$ and $Q_1 = X^5 + 17$. The identity $3H_1'Q_1 + H_1Q_1' = 10$ proves étaleness away from $w = 0$; there w is a parameter and H_1 a unit. The pole orders of X, w, T are $3, 5, 23$, and the different has degree 52. The wild inertia is $23{:}m$ with $m \mid 11$. If $m = 1$, the quotient different would be divisible by 22. Thus $m = 11$, and the formula $22 + 2h$ for that different gives the stated jumps.

On $Z_0 = \mathbf{P}^1_a$ set

$$\omega = -\frac{a^6}{a^{22} - 1}\, da, \qquad x = \frac{2a^{11}}{a^{11} - 1}. \tag{2}$$

The residue at $c \in \mathbf{F}_{23}^\times$ is c^7. Consequently $\omega = d \log \prod_c (a - c)^{[c^7]}$, where the brackets choose integer representatives between 1 and 22. If $\chi : H \to \mathbf{F}_{23}^\times$ is conjugation on P, let H act on a through χ^8. Since $7 \cdot 8 \equiv 1 \pmod{11}$, $h^*\omega = \chi(h)\omega$. The ramification data (m, h) at $x = 0, 1, \infty, 2$ are

$$(11, 7), \quad (1, 0), \quad (1, 0), \quad (11, 15).$$

There are exactly three ratios $h/m < 1$, none equal to 1, and all are less than 2.

Choose a wild point of each normal closure, identify its inertia with D, and its tame complement with H. Attach these two full G-tails to the induced central curve $\mathrm{Ind}_D^G(Y_0)$, where $Y_0 \to Z_0$ is purely inseparable of degree 23. The matched inertia groups and conductors, together with generation of G by the connected tail groups, verify the special G-deformation datum conditions of [22, Definition 2.13]. This constructs a connected special G-map, to which [22, Theorem 4.5] applies.

It is useful to retain the inseparable map explicitly. If $a = b^{23}$ on Y_0 and $z = b^{11}$ on $N = Y_0/H$, then

$$x = \frac{2z^{23}}{z^{23} - 1}, \qquad \overline{\beta} = \frac{1}{1 - x} = \frac{1 - z^{23}}{1 + z^{23}}. \tag{3}$$

The primitive attachment, new attachment, and two wild marks are $e = 0, q = \infty, A = 1, B = -1$. Their harmonic position is therefore part of the constructed special map. Replacing $a = b^{23}$ by $a = b$ would give the wrong quotient map.

The residues in the two pole orbits $a^{11} = 1, -1$ have opposite quadratic characters. They give the opposite classes $23A, 23B$ in the lift. A common change of wild generator rescales both classes together; taking inverses preserves square class. The tame tail supplies the class $2A$.

Proposition 5.2 (Descent of the marked lift)**.** *After ordering the two wild branches, the constructed special map has one isomorphism class of lifts over $\overline{K}$, and this class has a G-cover representative over K. Its degree-23 quotient has two individually K-rational totally ramified points A, B.*

Proof. We compute the automorphisms of this particular special map directly. On D_7, a pointed source automorphism is $X \mapsto aX + b$. It preserves the eight critical points, whose sum is zero, so $b = 0$. Comparison of coefficients gives $a^{14} = a^{21} = 1$, or $a^7 = 1$. All seven scalings preserve the map up to target scaling.

On the smooth completion of $Y^3 = X^5 + \lambda$, $\lambda \neq 0$, let q be infinity. Since $L(3q) = \langle 1, X \rangle$ and $L(5q) = \langle 1, X, Y \rangle$, a pointed automorphism has the form

$$X \longmapsto aX + b, \qquad Y \longmapsto cY + dX + e.$$

The Y^2 and X^4 coefficients force $d = e = b = 0$, and then $a^5 = c^3 = 1$. These fifteen automorphisms preserve $Y(X^6 + 15\lambda X)$ up to scaling. Their action on the parameter Y/X^2 at q is faithful.

The group facts $\mathrm{Out}(G) = 1$, $N_G(M) = M$, and $N_G(D) = D$ identify these quotient symmetries with the pointed G-equivariant tail symmetries. Indeed, a quotient symmetry lifts to the normal closure; correct its action on G by an inner automorphism. The correcting element normalizes M, so does not change the quotient symmetry. The point with stabilizer D is unique in its orbit, hence is fixed.

A G-equivariant automorphism of the special map preserves the chosen central component, and its restriction there over the original target lies in H. Such an element fixes both attaching points. Each pointed tail symmetry fixes all their G-translates. Thus these choices glue independently, and restriction to the normalization recovers them. The special-map automorphism group has order

$$|H| \cdot 7 \cdot 15 = 1155.$$

No matching of derivatives is required to glue automorphisms of the nodal special fibre $xy = 0$. This argument does not use a general automorphism-count formula for special G-maps.

By [22, Proposition 4.4 and Theorem 4.5], the lifts with identified special fibre form one tame inertia orbit of size $22 \cdot 7 \cdot 15 = 2310$. Forgetting that identification divides by the group just computed, freely: a stabilizer would give a generic G-equivariant automorphism over the target, belonging to $Z(G) = 1$. There are two unframed classes, exchanged by the quadratic cyclotomic action. Fixing the ordering of $23A, 23B$ leaves one class over K. The G-equivariant isomorphisms with its conjugates are unique, so satisfy the descent cocycle condition. They descend the cover, not merely its moduli point. Taking the M-quotient gives the source over K. Each totally ramified fibre contains one geometric point, which is consequently K-rational. □

5.2. **The marked distance determines the tame character.** The next lemma explains why the abstract good-reduction curve is not the only information needed. Distances are logarithmic annulus lengths, measured with $v(\pi) = 1$. Equivalently, in a smooth residue disk the distance to the disk separating two sections is the valuation of their parameter difference.

Lemma 5.3 (Marked distance and tame descent). *Let K' be a complete discretely valued field with algebraically closed residue field of characteristic p. Let C/K' be a smooth proper curve of genus at least two with potentially good reduction over a tame extension. Suppose that distinct K'-points A, B reduce to q on the good geometric special fibre D', and that $\mathrm{Aut}(D', q)$ is cyclic of order $n > 1$, prime to p, with faithful tangent character. If the marked separating vertex has distance $1/n$ from the good-reduction vertex, then the minimal good-reduction extension has degree n. Writing it as $K'(\rho)$ with $\rho^n = \pi$, the action $\tau(\rho) = \zeta\rho$ has tangent character ζ at q.*

Proof. Over a finite tame Galois extension with good reduction, the descent action extends by uniqueness of the stable model. Its kernel Δ on D' can be removed without losing smoothness. At any special point the completed ring is $R'[[u]]$. Averaging u over Δ gives an invariant parameter u_0 with the same reduction. Thus the invariant ring is $R'^{\Delta}[[u_0]]$, by coefficientwise invariance. The quotient of the proper projective model exists and is smooth by these local descriptions. Minimality therefore makes the residual action faithful; its order m divides n.

For an integral parameter ξ at q, the marked distance says $v(\xi(A) - \xi(B)) = 1/n$. Both values lie in the degree-m good-reduction field, whose value group is $\frac{1}{m}\mathbf{Z}$. Hence $n \mid m$, so $m = n$. This difference valuation is unchanged by an integral parameter change: the quotient of the two differences is its unit linear coefficient plus terms in the maximal ideal.

Tameness and the algebraically closed residue field give $K'(\rho)$ as stated. First linearize a special parameter and then lift it using the character projector $n^{-1}\sum_j \zeta^{-ej}\tau^j$. This projector is linear over the base valuation ring, not over the extension ring; its reduction is still a parameter. We obtain $\tau(\xi) = \zeta^e \xi$ exactly. Since the sections are rational, their values belong to $\rho^e \mathcal{O}_{K'}$, for $0 \le e < n$. The nonzero difference has valuation e/n plus an integer. Equality to $1/n$ forces $e = 1$. $\square$

In our application $23 \nmid |M|$, so the quotient of the semistable G-model by M commutes with reduction. Since D is transitive on the 23 letters, the quotient has one central component and one node at each attachment. Its three components form a chain

$$D_7 \text{ (rational)} \quad -- \quad N \text{ (marked rational)} \quad -- \quad D_{15} \text{ (genus four)}.$$

Forgetting the map and markings contracts the rational tree and gives good reduction D_{15}. The target annulus depths are $23/(2h)$ in v_{23} units. Each source annulus maps with degree 23; its length is the target length divided by 23 [23, Corollary 2.6]. In our valuation the two source lengths are therefore $1/7$ and $1/15$. Dividing by 253, the order of the inertia in the Galois closure, would be incorrect for this quotient map.

The pointed automorphism calculation above and lemma 5.3 give the good-reduction field $K(\rho)$, $\rho^{15} = \pi$. On $Y^3 = X^5 + \lambda$ its descent action is

$$\tau(X) = \zeta^{-3} X, \qquad \tau(Y) = \zeta^{-5} Y. \tag{4}$$

Indeed this is the unique pointed automorphism whose character on Y/X^2 is ζ. The hypotheses of the lemma are essential: neither an unmarked good-reduction curve nor a wild extension would justify this argument.

5.3. **A canonical model from differentials.**

Lemma 5.4 (The descended canonical lattice). *The marked quotient of proposition 5.2 has a flat canonical complete-intersection model over $\mathcal{O}$ with regular total space. Its closed fibre is rational with one cusp of value semigroup $\langle 3, 5\rangle$. In suitable closed coordinates it is*

$$HW - U^2 - \alpha UV - \beta V^2 = 0, \qquad UW^2 - V^3 = 0.$$

Its affine normalization is $(H, U, V) = (z^6 + \alpha z^4 + \beta z^2, z^3, z)$.

Proof. On the good special fibre the canonical forms

$$h = \frac{dX}{Y^2}, \quad u = Xh, \quad v = Yh, \quad w = X^2 h$$

have vanishing orders $6, 3, 1, 0$ at infinity and characters $7, 4, 2, 1$ modulo 15. The relative dualizing sheaf is invertible and commutes with base change [19, Tag 0E6N]. Lift these forms by cohomology and base change, and apply the tame character projectors. Their reductions form a basis, so their lifts do too. Multiplication in degrees two and three gives equivariant canonical equations q_g, c_g of characters $8, 6$ reducing to

$$hw - u^2 = 0, \qquad uw^2 - v^3 + \lambda h^3 = 0.$$

Surjectivity of these multiplication maps and the canonical Hilbert function identify the relative ideal with this quadric and cubic. Only canonical forms have been lifted. Lifting X, Y themselves with their special pole orders would impose extra Weierstrass conditions on the generic curve.

Set $(H, U, V, W) = (h, \rho^3 u, \rho^5 v, \rho^6 w)$. All four coordinates have character 7; dividing them all by ρ^7 gives an invariant generic basis. Thus the embedding is into $\mathbf{P}^3_K$, with no unexamined twist of the projective space. Its equations are

$$\begin{aligned} q &= \rho^6 q_g(H, \rho^{-3} U, \rho^{-5} V, \rho^{-6} W), \\ c &= \rho^{15} c_g(H, \rho^{-3} U, \rho^{-5} V, \rho^{-6} W). \end{aligned} \tag{5}$$

For a monomial of weight r, using weights $(0,3,5,6)$, the good-model coefficient has ρ-order congruent to $r-6$ for q and $r-15$ for c. If n_q, n_c are their least nonnegative residues, then n_q+6-r and n_c+15-r are nonnegative for all ten quadratic and twenty cubic monomials. This proves integrality of every coefficient. The possible closed equations are precisely

$$HW - U^2 + aUV + bUW + cV^2 + dVW + eW^2 = 0,$$
$$UW^2 - V^3 + fV^2W + gVW^2 + hW^3 = 0.$$

On $W=1$ put $V=z$ and solve for the monic cubic U and monic sextic H. Translate z to remove the quadratic term of U; linear changes remove its linear and constant terms and the cubic, linear and constant terms of H. The degree-five term of H vanishes with the same translation. This gives the asserted form.

The affine chart is $\mathbb{A}^1$. At infinity the coordinate orders are $3,5,6$, and the arithmetic genus of the complete intersection is four. Thus the unique singularity has $\delta=4$ and semigroup $\langle 3,5\rangle$, whose gaps are $1,2,4,7$. The closed equations form a regular sequence, giving flatness of the relative intersection; adjunction gives $\omega = \mathcal{O}(1)$. Finally the H^3 coefficient of c is π times a unit reducing to λ. At the cusp, on $H=1$, the linear parts of the two equations in the total local ring are $W + a_0\pi$ and $\lambda\pi$. They are independent. The total surface is regular there, and is relatively smooth at every other closed point. $\square$

In the good model's disk at q, the lifted ratio $\xi = v/w$ is an integral parameter of character one. The weighted coordinates give $z = \xi/\rho = V/W$; their other affine ratios are integral power series on the disk $|\xi| \le |\rho|$ and reduce to the cubic and sextic above. This is the marked central disk: $\xi(A) \in \rho\mathcal{O}$, and the separating radius is $|\rho|$ by lemma 5.3. Thus its reduction is N of equation (3), not an unrelated rational curve.

Proposition 5.5 (The effective divisor fixes the closed equations)**.** *Choose the harmonic coordinate on N with $e=0, q=\infty, A=1, B=-1$. The closure R_8 of the eight simple ramification points is finite flat of degree eight, with closed divisor $8e$ on the canonical model. The divided logarithmic differential is integral and satisfies*

$$\overline{\eta} = \overline{\frac{d\log\beta}{23}} = \frac{2z^8}{z^2-1}\,dz. \tag{6}$$

The closed equations can consequently be taken to be $q_0 = HW - U^2 - UV - V^2$ and $c_0 = UW^2 - V^3$ in this marked coordinate.

Proof. After a finite extension splitting the ramification points, all eight lie in the primitive disk and have z of positive valuation. The integral canonical ratios just constructed reduce at those points to their values at e. Properness and the valuative criterion therefore locate all their centers on the canonical model at e. This argument does not assume a contraction morphism from the semistable model to the canonical model.

Their schematic closure is proper with finite fibres, hence finite. It is torsion-free over $\mathcal{O}$, hence flat of degree eight. Near e the surface is smooth, so this is an effective Cartier divisor with closed equation z^8. Retaining this multiplicity is essential in the next step.

On the regular surface, $\operatorname{div}(\beta) = 23A - 23B$: there is no vertical contribution because the central reduction is nonconstant. Normality makes β a unit away from these disjoint sections and extends it to a morphism to $\mathbf{P}^1$. In particular there is no hidden base point at the cusp.

Take a primitive integral multiple s of $d\log\beta$ in $\omega(A+B)$. Its restriction to R_8 is generically zero and therefore zero, since the latter scheme is flat. Its closed-fibre restriction is nonzero: otherwise the multiplication-by-π exact sequence would contradict primitivity. On the normalization,

$$\nu^*\omega(A+B) = \omega_{\mathbf{P}^1}(8q + A + B)$$

has degree eight. Its section already vanishes at $8e$, so it has no further zero, and has nonzero residues at both marks. The residues of $d\log\beta$ are $23, -23$. Thus the primitive scaling divides by exactly 23, up to a unit. Normalizing the residue at A to one gives equation (6).

It remains to normalize the canonical equations without moving the four directions. Subtract $d\log((z-1)/(z+1))$ from the displayed differential. This leaves the dualizing differential

$$2(z^6+z^4+z^2+1)\,dz.$$

The subtracted differential is regular at q on the normalization, so belongs to the dualizing sheaf there. The canonical section W has no affine zero and is a constant multiple of dz. Using the canonical degrees $0,1,3,6$, write the two monic higher polynomials as $z^3+u_0z^2$ and $z^6+v_5z^5+v_4z^4+v_2z^2$ after removing lower basis terms. The quadric gives $v_5=2u_0$; the displayed differential then gives $u_0=v_5=0$ and $v_4=v_2=1$. This proves the stated equations in the harmonic coordinate, without simultaneously postulating an unjustified polynomial and marked-point normalization. □

5.4. **Comparing the two tails on the same sheets.** The first necessary deformation equations have two solutions. Only one is compatible with the common G-action. To compare them, one must transport the same wild generator through both nodes. The following elementary calculation makes that transport precise.

Lemma 5.6 (A leading coefficient at a node)**.** *Let S be a complete discrete valuation ring with uniformizer t, let $\alpha\in S$ have valuation $e>0$, and put $\mathcal{A}=S[[a,y]]/(ay-\alpha)$. Write v_C, v_D for the vertical valuations of the branches $y=0$ and $a=0$, normalized by $v_C(t)=v_D(t)=1$. Suppose $h\geq 2$, $F\in\mathcal{A}[1/t]$, and*

$$v_C(F)\geq he,\qquad v_D(F)\geq e,\qquad (F/\alpha^h)|_C=ca^{1-h}+O(a^{2-h}).$$

Then $(F/\alpha)|_D=cy^{h-1}+O(y^h)$. In particular, if σ preserves the node and its branches and $F=\sigma(a)-a$ satisfies the central condition, then

$$\sigma(y)-y=-cy^{h+1}+O(y^{h+2}).$$

Proof. The normal form in a, y, with a bounded t-denominator, gives a Laurent expansion $F=\sum f_na^n$ after substituting $y=\alpha/a$. The two valuation bounds say coefficientwise $v(f_n)\geq he$ and $v(f_n)+en\geq e$. There is no cancellation between distinct powers at either branch; only finitely many negative powers occur at a fixed t-order. The nth contribution to F/α on D is $f_n\alpha^{n-1}y^{-n}$. It vanishes for $n>1-h$ and has coefficient c for $n=1-h$. Lower indices give higher powers. For an automorphism, the second valuation bound follows from $v_D(\sigma(a))=v_D(a)=e$, and

$$\sigma(y)-y=-(F/\alpha)y\sigma(y)$$

gives the final assertion. □

We verify that the coordinates used here are actual node parameters. The primitive quotient disk has z-radius $|\pi|^{1/7}$. It is Galois invariant and contains a point over a tame extension: lift a smooth point of its component. Averaging the conjugates of its coordinate gives a K-rational center in the same disk. That center reduces to 0, hence lies in $\pi\mathcal{O}$ and is strictly inside the radius in question. The disk is therefore centered at 0. The primitive open annulus is $|\pi|^{1/7}<|z|<1$. The new open annulus is $|\rho|<|\xi|<1$, with coordinate $1/z=\rho/\xi$. Neither coordinate has a horizontal zero or pole in its annulus.

On a chosen G-node the quotient to the degree-23 source is by $H=C_{11}$. Tame averaging gives an eigenparameter x, with the opposite parameter of inverse character. The invariant node has parameters x^{11}, y^{11}. The pullback of z, or of $1/z$, has the same vertical divisor as x^{11} and no horizontal divisor through the node. Their ratio is a unit in the normal completed ring;

an eleventh root of that unit exists by Hensel's lemma. We may consequently choose actual parameters a_7, a_{15} with

$$a_7^{11} = z, \qquad a_{15}^{11} = 1/z.$$

These coordinates persist in the formal projective slice used below: a change $I + \pi M_0$, conjugated by the weights $(0, 3, 5, 6)$, changes the good coordinates only in ρ-orders at least 9.

Choose $r_c^{11} = \pi$, $s^7 = \pi$, and $\rho^{15} = \pi$ compatibly, and set

$$\alpha_7 = s^2 r_c^{-3}, \qquad \alpha_{15} = \rho^{-4} r_c^3.$$

Then $\alpha_7^{11} = s$, $\alpha_{15}^{11} = \rho$, and $\alpha_h^h = r_c$. Rescale the opposite node parameter to obtain $a_h y_h = \alpha_h$. This fixes the smoothing units, not just their valuations. The central residues of a_7, a_{15} can be chosen reciprocal.

At the generic point of N, the identities $d\beta = 23\beta\eta$ and equations (3) and (6) give the leading inverse-branch equation

$$(\delta z)^{22} = \pi^2 z^8 (1 - z^2)^{22}.$$

The coefficients of Taylor degrees 1 through 22 are divisible by 23; the degree-23 coefficient reduces to $-2/(1 + z^{23})^2$. After scaling, the residual equation is a unit times $v^{23} - v$, so its simple roots label the same P by $i \in \mathbf{F}_{23}$. Its action has leading term

$$\sigma_i(z) - z = r_c z^{4/11}(1 - z^2)i + \text{terms of larger central valuation.}$$

In the two node coordinates this becomes

$$\left.\frac{\sigma_i(a_7) - a_7}{r_c}\right|_C = \frac{i}{11} a_7^{-6}(1 - a_7^{22}), \qquad \left.\frac{\sigma_i(a_{15}) - a_{15}}{r_c}\right|_C = \frac{i}{11} a_{15}^{-14}(1 - a_{15}^{22}).$$

Apply lemma 5.6 to the regular functions $\sigma_i(a_h) - a_h$. It gives

$$\sigma_i(y_h) - y_h = -\frac{i}{11} y_h^{h+1} + O(y_h^{h+2}).$$

On the primitive tail $X = y_7^{-11}$, so $\sigma_i(X) - X = i y_7^{-4} + \cdots$. On the new tail put $\xi_{\text{tail}} = y_{15}^{-11} = X^2/Y$. Here

$$X = y_{15}^{-33} V_\lambda, \qquad V_\lambda^5 (V_\lambda - 1) = \lambda y_{15}^{165}, \quad V_\lambda(0) = 1,$$

so $\sigma_i(X) - X = 3i y_{15}^{-18} + \cdots$. Thus the Laurent root labels are i and $3i$ for the same wild element. This conclusion uses two vertical valuations of an actual node function; it is not obtained by exchanging limits in a central expansion. Eleventh-root ambiguities multiply labels by squares in $\mathbf{F}_{23}^\times$ and cannot exchange just one of the designs in the next calculation.

Exact computation 5.7 (The compatible first jet)**.** Trivialize the smooth quadric formally and use the nine-dimensional cubic slice specified in section 5.6. The first necessary function and logarithmic-differential equations have exactly two geometric solutions:

$$c = c_0 \pm \pi C_1 \pmod{\pi^2}, \qquad z_A = 1 + \pi \pmod{\pi^2}, \quad z_B = -1 + \pi \pmod{\pi^2},$$

where $C_1 = H^3 - 5HU^2 - 7HUV + 11HV^2$ and the sign of π is fixed by the first effective-fibre differential normalization $\eta_1 = (z + 15z^3 + 12z^5)\,dz \pmod{z^7}$. In the common labels above, only the positive solution is compatible with the two full M_{23}-tails.

Here is the finite test, including why its conclusion is exact. In the slice the residue, regularity, and differential conditions are a rank-18 linear system on 26 coefficients (the nine cubic coefficients, two marked motions, and fifteen numerator coefficients). On its eight-dimensional solution space, the next function equations have a degree-two residual ideal. Its reduced Gröbner basis is triangular with one equation $(t + 1)(t + 2) = 0$ and seven linear equations. Thus the two solutions above exhaust the geometric solutions, not just the $\mathbf{F}_{23}$-rational ones.

With the fixed translation subgroup P in S_{23} there are two containing conjugates of M_{23}: their number is $|N_{S_{23}}(P)|/|N_{M_{23}}(P)| = 22/11 = 2$. They preserve the two cyclic Witt designs $\mathcal{B}_0, \mathcal{B}_1$ on $\mathbf{F}_{23}$, each with 253 blocks and parameters $S(4, 7, 23)$. For reproducibility, $\mathcal{B}_0$ is given by the weight-seven words of the binary cyclic Golay code with generator $x^{11} + x^9 + x^7 + x^6 + x^5 + x + 1$; $\mathcal{B}_1$ is its image under $i \mapsto -i$. Exact permutation calculations verify their automorphism groups, their intersection 23:11, and generation of A_{23} by the two groups.

The first-jet equations give the primitive map $X^{23} + 5X^9 + 6X^2$ and the two candidate new-tail maps

$$Y^3 = X^5 + \varepsilon, \qquad T = 2Y(X^6 + 15\varepsilon X), \qquad \varepsilon = 1, -1.$$

These tails have the independently verified M_{23} groups of Exact computation 5.1, after scaling over k. For a design $\mathcal{B}$ form $I_{\mathcal{B}} = \sum_{B \in \mathcal{B}} \prod_{i \in B} X_i$. If the tail group preserves $\mathcal{B}$, this is a rational function of the target, integral on its affine line. Its pole bound at infinity is $7/23$ on the primitive tail and $21/23$ on the new tail, both less than one. It must therefore be constant.

The exact Laurent calculation excludes $\mathcal{B}_1$ for the primitive tail and for $\varepsilon = 1$, and excludes $\mathcal{B}_0$ for $\varepsilon = -1$. In the stored scaled series, the first nonzero remainders are respectively $15t^{1771}$, $5t^{759}$, and $5t^{759}$; the scaling and root equations are included in the checker. The computation solves and checks all 23 local roots to precision exceeding these exponents. A nonzero coefficient disproves constancy. Conversely, no finite zero expansion is used to prove constancy: the known M_{23} group and the exact two-design count identify the remaining design.

Both connected tails in one G-cover must act as the same subgroup on G/M. The negative candidate cannot do so, whereas the positive one agrees with the primitive tail. This proves the selection in Exact computation 5.7. In particular, the common sheet representation supplies information not contained in the two abstract group names or in the ramification conductors.

5.5. **From necessary jets to exact incidence.** The remaining local input is finite algebra. It is convenient to state exactly what is calculated before invoking the all-orders theorem.

Exact computation 5.8 (The required marked neighbourhood)**.** For the positive first jet, the necessary map and effective critical-fibre equations give

$$\begin{aligned} c &= c_0 + \pi C_1 + \pi^2 C_2 + \pi^3 (C_3 + v_0 U V^2) \pmod{\pi^4}, \\ z_A &= 1 + \pi + 9\pi^2 \pmod{\pi^3}, \\ z_B &= -1 + \pi - 9\pi^2 \pmod{\pi^3}, \end{aligned}$$

where v_0 remains an integral parameter and

$$\begin{aligned} C_2 &= 6H^2U + 5U^3 + 3H^2V + 4U^2V + 3UV^2, \\ C_3 &= -H^3 - 10HU^2 + 6HUV + 2HV^2. \end{aligned}$$

No osculating-incidence equation is imposed in obtaining these jets.

The calculation uses truncated canonical-ring multiplication, the two marked quintic-section systems, and the square of the effective degree-eight ramification factor. All first-order equation and marked-point directions are tested: the first variation of the normalized function is zero. This ensures that an omitted higher curve coefficient does not enter an earlier necessary equation. The first-jet ideal is computed with its degree-two identities; the next step leaves the UV^2 coefficient of C_2 free, and the effective square-fibre equations fix it to 3. They leave exactly the displayed v_0 direction at the following order. The scripts and independent residual-system audit are specified in section 11.

One normalization requires care. Let b denote the computational map normalized by $b(0) = 1$, and u its actual third branch value. Locally

$$b - u = R^2 F, \qquad \overline{R} = z^8, \qquad \overline{F} = z^7 \cdot \text{a unit}.$$

The vanishing first variation gives $u = 1 \pmod{\pi^2}$. The coefficient of z^{15} at order π then gives $R(0) \in \pi^2\mathcal{O}$. Since $F(0) \in \pi\mathcal{O}$,

$$u - 1 = -R(0)^2 F(0) \in \pi^5\mathcal{O}. \tag{7}$$

Thus replacing $b - u$ by $b - 1$ is legitimate in the earlier calculations modulo π^5, and constant scaling never changes $d \log b$. At all orders, however, u must be eliminated, not set to one. The fixed Hensel system below does exactly this. A finite-residue check finds valuation exactly five for $u - 1$ in the reconstructed lift, so the distinction is genuine.

We have now derived every hypothesis of theorem 5.9 for the lift of the constructed special map. The integral canonical equations come from lemma 5.4; the effective cluster and differential from proposition 5.5; and the required jets from the common-sheet comparison and Exact computation 5.8. The remaining theorem identifies any such lift with an independently verified exact model. Its proof uses necessary equations and uniqueness, not a claim that those equations generate the entire Hurwitz ideal. Together these results prove theorem 1.4.

5.6. **Exact reconstruction in a specified harmonic neighbourhood.** We isolate the local theorem used to complete the construction. Its low-order coefficients are explicit hypotheses, derived for our special map in the preceding subsections. The proof has two independent parts: necessary equations have at most one solution, and a small exact model supplies a solution. This separation is what permits an all-orders conclusion without extrapolating from finite precision.

Let $\mathcal{O}$ be the ring of integers of $K_{0,23} = \mathbf{Q}_{23}(\pi)$, where $\pi^2 = -23$. In canonical coordinates $[H : U : V : W]$ put

$$q_0 = HW - U^2 - UV - V^2, \qquad c_0 = UW^2 - V^3.$$

The normalization of their common zero curve in characteristic 23 has, on $W = 1$, the parametrization

$$(H, U, V) = (z^6 + z^4 + z^2, z^3, z).$$

The two marks are $z = 1, -1$; the critical cluster is at $z = 0$. In these coordinates the normalized reduced map is $(1 - z^{23})/(1 + z^{23})$. Thus $t_{\mathrm{n}} = (1 - z)/(1 + z)$ relates this description to proposition 4.4; the singular point $z = \infty$ has $t_{\mathrm{n}} = -1$.

Write

$$(m_0, \ldots, m_8) = (H^3, H^2U, HU^2, U^3, H^2V, HUV, U^2V, HV^2, UV^2)$$

and set

$$\begin{aligned} C_1 &= H^3 - 5HU^2 - 7HUV + 11HV^2, \\ C_2 &= 6H^2U + 5U^3 + 3H^2V + 4U^2V + 3UV^2, \\ C_3 &= -H^3 - 10HU^2 + 6HUV + 2HV^2. \end{aligned}$$

The neighbourhood considered below consists of equations and marks of the form

$$\begin{aligned} q &= q_0, \\ c &= c_0 + \pi C_1 + \pi^2 C_2 + \pi^3 C_3 + \pi^3 u_0 UV^2 + \pi^4 \sum_{i=0}^{7} u_{i+1} m_i, \\ z_A &= 1 + \pi + 9\pi^2 + \pi^3 u_9, \\ z_B &= -1 + \pi - 9\pi^2 + \pi^3 u_{10}, \qquad u_i \in \mathcal{O}. \end{aligned} \tag{8}$$

The remaining coordinates of the marks are determined on the smooth central chart by $q = c = 0$.

Theorem 5.9 (Incidence in the specified local neighbourhood)**.** *Suppose a smooth genus-*4 *curve and a degree-*23 *three-point map admit the canonical equations and ordered totally ramified marks in equation* (8)*. Suppose its remaining ramification consists of eight simple points in the formal neighbourhood of $z = 0$, and that the integral differential*

$$\eta = \frac{1}{23}\frac{d\beta}{\beta} \quad \textit{has reduction} \quad \overline{\eta} = \frac{2z^8}{z^2-1}\,dz, \qquad \operatorname{div}(\beta) = 23A - 23B.$$

Then the marked canonical curve is unique up to the coordinate normalizations just specified. It is isomorphic over $K_{0,23}$ to the exact model below. Its two osculating planes meet the curve in two distinct geometric points.

*The same conclusion holds over an unramified complete extension of $\mathcal{O}$ with the same congruences. The exact model has a degree-*23 *map with geometric monodromy M_{23} and represents the distinguished member of the seven-cover family.*

An intrinsic choice of coordinates. For $i = 0, 1, 2, 3$, the spaces

$$H^0(X, \omega_X(-iA - (3-i)B))$$

are one-dimensional in this neighbourhood, and their generators form a basis. This follows from rank-three jet minors and a nonzero basis determinant on the closed fibre. Use these generators as coordinates x_0, x_1, x_2, x_3. The marks become $[1 : 0 : 0 : 0]$ and $[0 : 0 : 0 : 1]$, with vanishing sequences $(0, 1, 2, 3)$ and $(3, 2, 1, 0)$. The quadric consequently has the form

$$ax_0x_3 + bx_0x_2 + dx_1x_2 + ex_1^2 + fx_1x_3 + gx_2^2.$$

Here a, b, d, e are units. Substituting $x_i = d_iy_i$ and multiplying the equation by κ, with

$$d_0 = 1, \qquad d_1 = b/d, \qquad \kappa = (ed_1^2)^{-1}, \qquad d_2 = (\kappa b)^{-1}, \qquad d_3 = (\kappa a)^{-1},$$

sets the first four displayed coefficients to one. This normalization is unique and requires no extension of the field. Reduce the cubic modulo the monic quadric in lexicographic order $x_0 > x_1 > x_2 > x_3$, and set its $x_0^2x_3$ coefficient to one. These operations remove the scalar choices in the four generators.

The exact model. In these normalized coordinates the quadric is

$$q_* = x_0x_2 + x_0x_3 + x_1^2 + x_1x_2 - \frac{127}{229}x_1x_3 + \frac{12192}{52441}x_2^2. \tag{9}$$

The cubic c_* has the following nonzero coefficients; all others vanish.

Monomial	Coefficient in c_*
$x_0^2x_3$	1
$x_0x_1x_3$	$(69 + 41\pi)/916$
$x_0x_3^2$	$-(288 + 3552\pi)/52441$
x_1^3	$-(5 + \pi)/458$
$x_1^2x_2$	$-(3061 + 569\pi)/209764$
$x_1^2x_3$	$-(8649 + 1515\pi)/104882$
$x_1x_2^2$	$-(298479 + 49803\pi)/48035956$
$x_1x_2x_3$	$-(1041970 + 181274\pi)/12008989$
$x_1x_3^2$	$(2318419 + 389759\pi)/48035956$
x_2^3	$-(2350008 + 338328\pi)/2750058481$
$x_2^2x_3$	$-(55642056 + 9354216\pi)/2750058481$

Proof of theorem 5.9. We separate local uniqueness, exact existence, and identification of the cover. In particular, recognizing coefficients from a local approximation is not used as a proof of their exact values.

Local uniqueness. The nine cubic directions m_i give a formal slice for canonical curves with quadric q_0: the six infinitesimal orthogonal-coordinate directions, cubic scaling, the four multiples of q_0, and these nine directions span the twenty-dimensional cubic space. The determinant is 10 in $\mathbf{F}_{23}$ for the certificate's basis. The formal inverse-function theorem therefore justifies this slice.

Choose bases S_A, S_B of the four-dimensional kernels of the order-23 quintic jet maps. Fixed unit minors make their coefficients analytic functions of the eleven parameters u_i. An invertible minor of fifteen cross-product equations determines a multiplier matrix M, with one entry fixed to remove its scalar. It defines a local ratio N/D, normalized to value one at $z = 0$, even off the locus where all global cross-products vanish. On a genuine cover this is its function β, up to scalar. No global map is inferred merely from solving those fifteen rows.

Let $R(z)$ be the degree-eight Weierstrass factor of η, with $R \equiv z^8 \pmod{\pi}$. Reduce $N - D$ and D modulo R^2, obtaining P, Q, and put

$$E = P - \frac{P(0)}{Q(0)}Q, \qquad E_j = [z^j]E.$$

Here $Q(0)$ is a unit. Retaining R^2 retains the effective ramification multiplicities. For a three-point map $E = 0$, because its eight simple critical points have one common value. Let C_j denote the central-series coefficient of z^j in the negative of the first remaining cross-product $S_{A,0}(MS_B)_1 - S_{A,1}(MS_B)_0$, using the fixed bases in the certificate. Eleven necessary equations are

$$F = \left(\frac{E_{15}}{\pi^3}, \frac{E_9}{\pi^4}, \ldots, \frac{E_{14}}{\pi^4}, \frac{C_7}{\pi^5}, \ldots, \frac{C_{10}}{\pi^5}\right) = 0. \tag{10}$$

These are integral restricted power series on the stated neighbourhood. The exact residual calculation gives

$$\overline{F} = J\overline{u} + b, \qquad \det J = 16 \in \mathbf{F}_{23}^{\times}, \qquad -J^{-1}b = (0, 0, -7, 0, -1, -2, 0, 10, 0, -7, -7).$$

The full matrix, fixed jet bases, and normalizations are retained in the local certificate and its accompanying derivation.

This calculation is an identity test, not a sample of possible covers. The parameters first enter at orders π^3, π^4. The first variation of the normalized ratio vanishes modulo π; its variable contribution begins at order π^4. After division by $23 = -\pi^2$, the coefficients of R first vary at order π^2. Modulo the precisions needed in equation (10), all expressions have parameter degree at most two. Values at zero, the positive and negative coordinate vectors, and the sums of two distinct coordinate vectors therefore determine them. The 78 evaluations verify integrality of the divisions and the displayed affine reduction. The computation also verifies that the two unavailable final digits of the Weierstrass factor do not affect these divided reductions.

Consequently $F(u) = \widetilde{J}u + \widetilde{b} + \pi G(u)$, with G integral and restricted and $\widetilde{J}$ invertible over $\mathcal{O}$. The map $u \mapsto -\widetilde{J}^{-1}\widetilde{b} - \pi\widetilde{J}^{-1}G(u)$ is a strict contraction on $\mathcal{O}^{11}$. It has a unique fixed point, also after the stated base extensions. Every cover in the theorem satisfies these necessary equations. We do not need to assert that the selected equations alone imply all the other map equations.

Exact existence. The pair (q_*, c_*) was recovered from the local lift in the intrinsic coordinates above, without using the previously stored generic models. The following independent tests are over K_0. The Jacobian-minor ideal is the unit ideal on all four affine projective charts, so the curve is a smooth genus-4 complete intersection. Its marked vanishing sequences are as stated. The order-23 quintic jet kernels each have dimension four. Their common base divisors are exactly

$23A$ and $23B$: saturation excludes other base points, and a section has order exactly 23 at each mark.

In the degree-ten canonical ring, of dimension 57, an exact rank-15 multiplier calculation and all six cross-product identities give a function with divisor $23A - 23B$. Away from its zero and pole, a critical ideal in the chart $x_1 = 1$ has an eight-dimensional reduced quotient algebra, and $N = uD$ in that algebra for one nonzero scalar u. Reducedness is certified by a squarefree degree-eight characteristic polynomial of a multiplication operator. Riemann–Hurwitz gives total ramification 52; the two marked points contribute 44, and the eight distinct critical points exhaust the remainder. Thus each is simple, there are no further critical points in another chart, and the map has the required three-point passport.

An integral projective change with unit determinant identifies this exact marked model with the stated local slice modulo π^{33}. Correcting the quadric and cubic by the formal slice construction preserves the lower-order congruences in equation (8). The normalized map is determined by its divisor; its divided differential and critical cluster are therefore those used to form the necessary equations above. The exact map belongs to the required neighbourhood and hence, by uniqueness, is the local lift to all orders.

Incidence and identification. The two osculating planes on this exact model are $x_3 = 0$ and $x_0 = 0$. On their common line the identity

$$c_*(0, x_1, x_2, 0) = \left(x_1^2 + x_1x_2 + \frac{12192}{52441}x_2^2\right) \cdot \left(-\frac{5+\pi}{458}x_1 - \frac{771+111\pi}{209764}x_2\right)$$

has first factor equal to $q_*|_{x_0=x_3=0}$. Its discriminant is $3673/52441 \neq 0$. This proves the claimed two reduced points, exactly rather than to finite precision.

Finally, a separate comparison certificate constructs an invertible projective matrix between this model and the degree-one model of section 3. It sends the ordered marks to the ordered marks and verifies equality of the two rational functions up to a nonzero target scalar by exact reduction in the canonical ring. The existing certified branch cycles consequently give geometric monodromy M_{23} for this map. This is an exact isomorphism argument, not an inference from the passport and not a new numerical continuation calculation. □

Remark 5.10 (An explicit descent of the reconstructed map)*.* The exact model also admits a direct descent, independently of the exclusion of incidence on the other six covers. Put $a = -127/229$, $b = 12192/52441$, and define

$$T(x_0, x_1, x_2, x_3) = (abx_3, bx_2, x_1, x_0/a).$$

If $\sigma(\pi) = -\pi$, exact substitution gives

$$T^2 = bI, \qquad q_*(Tx) = bq_*(x), \qquad \sigma(c_*)(Tx) \equiv \mu c_*(x) \pmod{q_*},$$

where $\mu = -(52130760 + 33951768\pi)/1525141603$ and $\mu\sigma(\mu) = b^3$. Thus $R = (b/\mu)T$ satisfies $R\sigma(R) = I$. Solving $v = R\sigma(v)$ gives a four-dimensional $\mathbf{Q}$-space and an invertible change of coordinates in which the curve equations are rational; the certificate checks these equations directly.

For the map f normalized to have third branch value 1, one also has $f(x)\sigma(f)(Tx) = 1$ in the function field. Consequently $t = (f-1)/(\pi(f+1))$ is fixed by the descent and defines a map over $\mathbf{Q}$. Its branch locus is $t = 0$ and $23t^2 + 1 = 0$. Rescaling by $T_{\text{base}} = 23t$ gives the branch locus $0, \pm\sqrt{-23}$ used in the paper. This recovers the known rational cover in different equations.

Remark 5.11 (Why uniqueness is enough). The eleven equations are necessary for a cover in this neighbourhood. We do not assert that their ideal is the entire map ideal. Nor do the displayed low-order incidence cancellations prove an exact incidence relation on their own: an off-locus audit detects a higher defect in the constant residual multiplier. What closes the proof is that the constructed lift and the independently verified exact model both satisfy the necessary system, which has at most one solution. Their identification transfers the exact incidence calculation to the lift. The earlier coefficient checks place the model in the neighbourhood; uniqueness, not the number of matching digits, supplies equality to all orders.

5.7. **What the construction explains.** The special map supplies the harmonic configuration, but the characteristic-zero incidence needs more. The marked distance fixes the descent action on the good curve. Its canonical lattice and effective ramification divisor fix the closed canonical geometry. The common Mathieu action excludes the other first jet. Finally, unit-Jacobian uniqueness and exact existence identify the entire lift. Each step retains information which the preceding, coarser description would lose.

There are also two distinct uniqueness statements. Local uniqueness identifies a lift in this neighbourhood; it does not assert that all covers with the passport enter this neighbourhood. Global uniqueness of the incidence locus is proved on all seven exact models in Exact computation 2.2. Its intrinsic definition then gives Galois fixedness by proposition 2.3.

Accordingly, the construction explains the exceptional incidence for an explicit harmonic special M_{23}-map. It remains computational at the tail-group, jet, and exact-model stages. It is not a permutations-only existence argument, nor a general theorem that harmonic reduction forces rationality. The finite Fano–affine comparison considered next is a further question, not an unstated ingredient of this proof.

6. Finite invariants and the component comparison

We return to the exact seven-point algebra. The invariants in this section give a further description of its two components. Their comparison uses the certified identification of the seven covers; it is not needed to construct the harmonic lift.

6.1. **Descent of finite invariants.** We first specify what *descent* means here. For a finite set S, let $\underline{S}_{K_0} = \coprod_{s\in S} \operatorname{Spec} K_0$ be the associated constant finite étale scheme. An assignment

$$f : \mathcal{H}_{\mathrm{in}}(\overline{K}_0) \longrightarrow S$$

descends over K_0 if it is invariant under $\operatorname{Gal}(\overline{K}_0/K_0)$, equivalently if it is induced by a K_0-morphism $\mathcal{H}_{\mathrm{in}} \to \underline{S}_{K_0}$. When $S = \{0,1\}$, such morphisms are equivalent to idempotents of the coordinate algebra A_{H}.

We next define the finite-group assignments to be tested. Let $G = M_{23}$, let (x, y, z) be one of the seven normalized triples, and put $Y = \langle y\rangle$ and $Z = \langle z\rangle$. The two-endpoint order count is

$$\nu(Y, Z) = \#\big\{(a, b) \in (\mathbf{F}_{23}^{\times})^2 : \operatorname{ord}(y^a z^b) = 23\big\}.$$

For $c \in \{y, z\}$ and $d \in \mathbf{F}_{23}^{\times}$, put

$$a_c(d) = \operatorname{ord}\big(xx^{c^d}\big), \qquad \kappa_m = \sum_{c\in\{y,z\}} \#\{d \in \mathbf{F}_{23}^{\times} : a_c(d) = m\} \quad (m = 2, 4).$$

Changing a generator of Y or Z only reindexes the exponents, and the sum over $c \in \{y, z\}$ is unchanged when the two endpoints are exchanged.

Write $N(Y) = N_G(Y)$, use the convention $x^g = g^{-1}xg$, and set

$$\Omega = x^{N(Y)} \cap x^{N(Z)}.$$

The stabilizer of x in either normalizer is trivial: its order divides both $|N_G(Y)| = 253$ and $|C_G(x)| = 2688$. Thus, for $w \in \Omega$, the transporter sets from x to w in $N(Y)$ and $N(Z)$ are singletons; denote their elements by $n_Y(w)$ and $n_Z(w)$. Their relative comparison

$$c_w = n_Y(w)n_Z(w)^{-1}$$

centralizes x. Let A_x be the set of eight two-element orbits of $\langle x\rangle$ on the moved letters. The induced permutation $\overline{c}_w$ lies in

$$C_G(x)/\langle x\rangle \cong 2^3{:}\,\mathrm{PSL}_3(2) \cong \mathrm{AGL}(3,2),$$

acting faithfully on A_x. Define

$$\mu = \#\{\overline{c}_w : w \in \Omega\}, \qquad \Theta = \sum_{w\in\Omega} [\overline{c}_w] \in \mathbf{F}_2[C_G(x)/\langle x\rangle].$$

Finally, let ϵ be the standard augmentation homomorphism of this group algebra. Then $\epsilon(\Theta) = |\Omega| \bmod 2$. The Hamming weight $\mathrm{wt}(\Theta)$ is the number of nonzero coefficients of Θ in the group basis.

Exact computation 6.1 (Descent of the transporter invariants)**.** In the sextic embedding order $(7,4,1,5,3,2)$, the exact values of the two-endpoint order count ν, the affine-image count μ, and the Hamming weight of Θ are respectively

$$\begin{aligned}\nu|_L &= (42,46,54,46,28,54),\\ \mu|_L &= (14,28,16,28,31,16),\\ \mathrm{wt}(\Theta)|_L &= (11,27,15,27,25,15).\end{aligned}$$

The cyclic-conjugacy order counts (κ_2,κ_4) are likewise

$$((4,12),(2,6),(0,16),(2,6),(0,8),(0,16)).$$

None is constant. By contrast,

$$\epsilon(\Theta)|_L = (1,1,1,1,1,1), \qquad \epsilon(\Theta)|_{K_0} = 0.$$

Moreover, on all seven geometric points,

$$\mathbf{1}_{\{\nu=32\}} = \mathbf{1}_{\{\mu=17\}} = \mathbf{1}_{\{(\kappa_2,\kappa_4)=(0,12)\}} = \mathbf{1}_{\{\epsilon(\Theta)=0\}}.$$

Proposition 6.2 (Component idempotents detected by finite invariants)**.** *Let $e_1 = (1,0)$ and $e_6 = (0,1)$ be the primitive idempotents of $A_{\mathrm{H}} = K_0 \times L$. The four $\{0,1\}$-valued assignments in Exact computation 6.1 descend to morphisms to $\underline{\{0,1\}}_{K_0}$ and all correspond to e_1; the augmentation corresponds to $e_6 = 1 - e_1$.*

The assignments ν, μ, and (κ_2,κ_4) do not descend on $\operatorname{Spec} L$ *to morphisms into the corresponding constant finite schemes. Moreover Θ cannot descend as a section of a lisse $\mathbf{F}_2$-sheaf whose transition maps permute the natural group-algebra basis.*

Proof. Because L is a field, $\operatorname{Spec} L$ is connected. Every morphism from it to a constant finite étale scheme is therefore constant, whereas the displayed values of the three assignments vary among its six geometric points. A permutation of the group-algebra basis preserves Hamming weight, so the varying weights also rule out the stated basis-compatible descent of Θ. The four equality predicates are 1 only at ID 6, while the augmentation is 1 precisely on the other six points. Under the correspondence between morphisms to the constant two-point scheme and idempotents, they are therefore e_1 and e_6, respectively. $\square$

This interpretation uses the certified $1+6$ decomposition and the exact branch-cycle identification of its seven geometric points.

Proof of theorem 1.3. By proposition 2.3, the nonempty osculating-incidence locus has characteristic function $e_{\mathrm{osc}} = e_1$. By proposition 4.3, evaluation at the distinguished singularity defines $e_{\mathrm{sing}} = e_6$. By proposition 6.2, the augmentation is $\epsilon(\Theta) = e_6$. Thus $1 - e_{\mathrm{osc}} = e_{\mathrm{sing}} = \epsilon(\Theta)$ in A_{H}. □

6.2. **Frobenius and unramified extensions.** We give a coordinate-independent interpretation of the local idempotent. The general facts used here are standard properties of finite fields and henselian rings; the application identifies what the singular-position calculation measures in this particular Hurwitz scheme.

Theorem 6.3 (Local component criterion)**.** *Let R be the ring of integers of a nonarchimedean local field F, with residue field $k = \mathbf{F}_q$. Let A be a finite normal flat R-algebra with étale generic fibre, and write*

$$A = \prod_i O_i, \qquad B = (A/\mathfrak{m}_R A)_{\mathrm{red}} = \prod_i \mathbf{F}_{q^{f_i}}.$$

Here O_i is the ring of integers of a field extension of F with ramification index e_i and residue degree f_i.

(1) *The scheme-theoretic fixed locus of the q-power Frobenius on $\operatorname{Spec} B$ is open and closed, consisting precisely of the factors with $f_i = 1$. It lifts uniquely to an open-and-closed subscheme $Z \subset \operatorname{Spec} A$, finite flat over R of rank $\sum_{f_i=1} e_i$. In particular, $Z \simeq \operatorname{Spec} R$ if and only if there is exactly one index with $f_i = 1$ and that index has $e_i = 1$.*

(2) *Let R_m/R be the unramified extension of degree $m \geq 1$. On the connected component $\operatorname{Spec} O_i$, its pullback is the split degree-m cover if and only if $m \mid f_i$. Thus the union of the components on which it splits is canonically open and closed.*

These constructions are invariant under R-algebra isomorphisms of A.

Proof. The fixed locus is the equalizer of Frobenius and the identity, so its coordinate ring is

$$B/(b^q - b : b \in B).$$

In a field factor $\mathbf{F}_{q^{f_i}}$ this ideal is zero if $f_i = 1$. Otherwise there is an element not fixed by q-power Frobenius; its nonzero difference is a unit, so that factor of the quotient is zero. This proves the description of the fixed locus. Notice that it is a quotient algebra, not the invariant subalgebra B^{Frob_q}.

Idempotents lift uniquely through nilpotent ideals and from the closed fibre of a finite algebra over a complete local ring [19, Tags 09XD and 04GK]. This proves existence and uniqueness of Z. Each O_i is free over R of rank $e_i f_i$, which gives the stated rank. A factor of rank one has fraction field F and hence is R.

The residue algebra of $O_i \otimes_R R_m$ at the maximal ideal of O_i is $\mathbf{F}_{q^{f_i}} \otimes_{\mathbf{F}_q} \mathbf{F}_{q^m}$. As an algebra over $\mathbf{F}_{q^{f_i}}$, this is a product of $\gcd(f_i, m)$ fields, each of degree $m/\gcd(f_i, m)$. It is therefore split exactly when $m \mid f_i$. Finite étale algebras over the henselian local ring O_i are determined by their residue algebras [19, Tag 04GK], so the same criterion holds over O_i. The descriptions use neither coordinates nor labels of the factors, proving the last assertion. □

The two criteria in the theorem are not equivalent in general. For example, an odd residue degree greater than one gives neither a Frobenius-fixed point nor splitting of the unramified quadratic extension. They are complementary when every residue degree is either one or two.

Corollary 6.4 (The characteristic-23 component). *Let $R = \widehat{\mathcal{O}}_{K_0,\mathfrak{q}}$, where $\mathfrak{q}$ is the prime above 23, and let A be the normalization of R in $A_{\mathrm{H}} \otimes_{K_0} \operatorname{Frac} R$. The complement of the Frobenius-fixed locus on its reduced closed fibre lifts to the component idempotent e_6. Equivalently, e_6 is the idempotent of the locus on which the finite étale double cover*

$$\operatorname{Spec}\big(A[z]/(z^2+1)\big) \longrightarrow \operatorname{Spec} A$$

splits. In the singular-position coordinate of proposition 4.3, its reduction is

$$e_{\mathrm{sing}} = -(u^{23}-u)^{22} = 2u^4 + 2u^3 + 4u^2 + 2u + 3 \quad \text{in } \mathbf{F}_{23}[u]/(R_{23}).$$

Proof. The exact local decomposition has pairs $(e_i, f_i) = (1,1), (1,2), (2,2)$. The first factor comes from K_0 and the other two from L. Thus theorem 6.3 identifies the Frobenius-fixed component with the R factor and its complement with the localization of L.

Since -1 is not a square in $\mathbf{F}_{23}$, $R[z]/(z^2+1)$ is the unramified quadratic extension of R. Its discriminant is a unit. It is nonsplit on the first factor and split on the other two, by the same theorem.

Finally, let $\delta = u^{23} - u$. It is zero in the linear factor. In each quadratic factor u generates the residue field, so $\delta \neq 0$ and $\delta^{23} = -\delta$. Consequently $\delta^{22} = -1$ there. Thus $-\delta^{22}$ has values $(0,1,1)$, as does the displayed polynomial. The Chinese remainder theorem proves their equality. □

This gives an actual finite étale cover of the normalized integral Hurwitz scheme, not just a list of three matching values. Its split locus is defined without the chosen coordinate u. The singular-position polynomial is an explicit presentation of that locus. Combining corollary 6.4 with the certified branch-cycle matching gives

$$\epsilon(\Theta) = e_6 = e_{\mathrm{sing}}.$$

The construction of the local locus does not use that matching; the comparison with $\epsilon(\Theta)$ still does.

There are two essential limitations. First, the residue field is part of the statement: replacing q-power Frobenius by q^d-power Frobenius fixes the factors with $f_i \mid d$, and can change the locus. Second, a unique local rank-one factor need not descend to a rational point over a number field. For example, $X^3 - 2$ is irreducible over $\mathbf{Q}$, whereas modulo 5 it factors as

$$(X-3)(X^2+3X+4),$$

with the quadratic factor irreducible and the two factors coprime. Its algebra over $\mathbf{Q}_5$ has one rank-one factor and one unramified quadratic factor, but $\operatorname{Spec} \mathbf{Q}(\sqrt[3]{2})$ has no $\mathbf{Q}$-rational point. In the present application it is the already proved global decomposition $A_{\mathrm{H}} = K_0 \times L$ that supplies descent. The local criterion alone is not a theorem giving regular Galois realizations.

6.3. **Finite incidence and the unresolved geometric comparison.** The relative osculating incidence of proposition 2.3 is already constructed in characteristic zero. The separate question here concerns specialization of the Fano–affine finite incidences below. These identities specify the finite data, but do not extend them to relative correspondences on a semistable M_{23}-cover. Fix the natural degree-23 permutation representation $M_{23} \subset S_{23}$. Let $x \in 2A(M_{23})$, let $H_x = \operatorname{Fix}(x)$, and let A_x be the set of its eight two-element sheet orbits. Thus H_x, with its natural incidence relation, is the seven-point Fano plane, and A_x is an affine torsor under $\mathbf{F}_2^3$. Put $B_x = H_x \amalg A_x$.

Lemma 6.5 (Fano–affine odd fixed-point lemma). *For every $c \in C_{M_{23}}(x)$,*

$$|\operatorname{Fix}_{H_x}(c)| + |\operatorname{Fix}_{A_x}(c)| \equiv 1 \pmod 2.$$

Proof. Write $U = O_2(C_{M_{23}}(x))/\langle x\rangle \simeq \mathbf{F}_2^3$. The centralizer acts on H_x through $\mathrm{GL}(U)$ acting on $U^* \setminus \{0\}$ and on A_x through $U{:}\,\mathrm{GL}(U)$. Write the image of c as an affine map (t,g). If $g-1$ is singular, its dual fixes a nonzero space of dimension d, so it fixes $2^d - 1$ points of H_x, an odd number. The affine fixed-point equation has either no solution or an affine space parallel to $\ker(g-1)$, hence an even number of solutions when nonempty. If $g-1$ is invertible, the dual action fixes no point of H_x, while the affine equation has exactly one solution. Both cases give the assertion. $\square$

After a splitting extension and a common identification of the geometric sheets, let $D_y, D_z \simeq 23{:}11$ be the normalizers associated with the two marked wild branches. For $D = D_y$ or D_z, define

$$\mathcal{L}_D = \{(b, x^d, b^d) : b \in B_x,\ d \in D\}.$$

The action of D on x^D is free because a stabilizer order divides both 253 and $|C_{M_{23}}(x)| = 2688$. If $w \in x^{D_y} \cap x^{D_z}$, let $d_y(w)$ and $d_z(w)$ be the unique transporters carrying x to w. Then $c_w = d_y(w)d_z(w)^{-1} \in C_{M_{23}}(x)$. Equality of the basis elements indexed by b in the two incidence correspondences over w is equivalent to $b^{c_w} = b$. The preceding lemma gives

$$\deg(\mathcal{L}_{D_y} \cap \mathcal{L}_{D_z}) \bmod 2 = |x^{D_y} \cap x^{D_z}| \bmod 2 = \epsilon(\Theta). \tag{11}$$

Equivalently, after the splitting extension, regard B_x and each $\mathcal{L}_D$ as finite *étale* schemes. Take constant $\mathbf{F}_2$-sheaves and use $\mathcal{L}_{D_y}$ and $\mathcal{L}_{D_z}$ as incidence correspondences in opposite directions. Their Lefschetz–Verdier pairing on the generic fibre is exactly the finite intersection in (11). Thus the generic construction specifies the schemes, sheaves, correspondences, and cohomological maps explicitly.

6.3.1. *The local nearby-cycle calculation.* Let R be a strictly henselian discrete valuation ring with residue field k and fraction field K, and let $p : E \to Q$ be a finite surjection. The finite Ferrand pushout [18]

$$\mathcal{P}_p = \operatorname{Spec}(R^E) \amalg_{\operatorname{Spec}(k^E)} \operatorname{Spec}(k^Q)$$

has coordinate ring

$$A_p = R^E \underset{k^E}{\times} k^Q = \{(a_e) \in R^E : \bar{a}_e \text{ is constant on each fibre of } p\}.$$

For $q \in Q$, the strict local geometric generic fibre is the discrete set $E_q = p^{-1}(q)$. Therefore, with $\Lambda = \mathbf{F}_2$,

$$(R^0\Psi\Lambda)_q = \Lambda[E_q], \qquad (R^i\Psi\Lambda)_q = 0 \quad (i > 0). \tag{12}$$

Let $\alpha, \beta \in \Lambda^E$ select the two endpoint incidences. The generic cohomological maps are the identity on the selected normalization sections and zero on the others. Their generic pairing, and the pairing of their nearby-cycle specializations, are both

$$\sum_{e \in E_q} \alpha_e \beta_e. \tag{13}$$

Indeed, the specialized maps are

$$\Lambda \longrightarrow \Lambda[E_q], \quad 1 \longmapsto \sum_e \alpha_e[e], \qquad \Lambda[E_q] \longrightarrow \Lambda, \quad [e] \longmapsto \beta_e.$$

This is the elementary finite-stalk instance of compatibility of the Lefschetz–Verdier pairing with nearby cycles in Lu–Zheng [24, Corollary 3.10].

There are also canonical maps

$$\Lambda \longrightarrow \Lambda[E_q] \longrightarrow \Lambda, \qquad 1 \longmapsto \sum_e [e], \quad \sum_e a_e[e] \longmapsto \sum_e a_e.$$

Inserting these maps and then applying augmentation produces the fibre-sum product

$$\left(\sum_e \alpha_e\right)\left(\sum_e \beta_e\right). \tag{14}$$

This is a different correspondence, not the specialization of (13). For $E_q = \{1, 2\}$, $\alpha = (1, 0)$, and $\beta = (0, 1)$, the basiswise pairing is 0 while the product after augmentation is 1. Thus a conductor quotient cannot by itself justify replacing (13) by (14).

The normalization–conductor triangle does not alter this conclusion. After restriction to the generic fibre its conductor-supported terms vanish; nearby cycles recover the full permutation module in (12), not a rank-one quotient. Applying the counit $p_*\Lambda_E \to \Lambda_Q$ is an additional operation and must be incorporated into the relative correspondence if it is to be used.

6.3.2. *What the integral multiplicities do and do not prove.* Put $\mathcal{C} = 2A(M_{23})$, $T_D = x^D$, and $P_D = \mathcal{C} \setminus T_D$. At this stage these are finite subsets of a permutation group, not effective cycles on a constructed relative curve. For the natural sheet representation ρ, exact counting gives

$$\rho(\mathcal{C}) = 120J + 1035I, \qquad \rho(T_D) = 8J + 69I, \qquad \rho(P_D) = 112J + 966I.$$

In particular, a diagonal coefficient of the last matrix is 1078 and an off-diagonal coefficient is 112. Their divided difference is

$$(1078 - 112)/2 = 483 \equiv 1 \pmod 2.$$

This is a calculation with specified integral representatives. The congruent representative $P_D - 2\mathcal{C}$ has image $-128J - 1104I \equiv 0 \pmod 4$. Hence, in the unrestricted mapping cone of the graph-to-sheet map, the mod-2 class represented by P_D has a mod-4 lift and its coefficient Bockstein is zero.

For a binary vector of even weight, the function

$$Q(v) = \mathrm{wt}(v)/2 \pmod 2$$

satisfies

$$Q(v + w) = Q(v) + Q(w) + \langle v, w\rangle.$$

Indeed, the integer weight of $v + w$ is the sum of the two weights minus twice the size of the intersection of their supports. Each T_D has odd size 253, so Q is not defined on its indicator alone. Its value on the even-weight symmetric difference is obtained directly from $|T_{D_y} \triangle T_{D_z}| = 506 - 2|T_{D_y} \cap T_{D_z}|$. Together with $|P_D| = 3542$, this gives

$$Q(T_{D_y} + T_{D_z}) = 1 + \epsilon(\Theta), \qquad Q(P_D) = 1.$$

These are finite quadratic identities. They do not by themselves give a quadratic refinement on nearby cycles.

In particular, the determinant of a perfect complex [20] is not automatically a symmetric line with degree equal to half its Euler characteristic. Assigning degree m mod 2 to the determinant of a rank-$2m$ module does not define a symmetric monoidal functor to ordinary graded lines: exchanging two rank-two blocks has determinant sign $+1$, whereas exchanging their proposed odd lines has Koszul sign -1. Additional categorical structure and coherence data would be needed.

A geometric comparison would require a relative model, cohomological correspondences extending both finite incidences, and a construction retaining the two specified integral multiplicity vectors $\mathcal{C}$ and T_D. It would then have to prove, rather than assume, the identification of its generic invariant with $\epsilon(\Theta)$ and its special invariant with e_{sing}. Neither the finite pinching calculation above nor ordinary nearby-cycle duality [24] supplies these identifications. We do not assert such a relative construction or a second proof of theorem 1.3 here.

The next two sections record supporting arithmetic on the distinguished cover: first its explicit minimal projection, then the Fano and affine arithmetic of its branch fibre.

7. The minimal-degree projection

Let $F(T,V) \in \mathbf{Z}[T,V]$ be the integral source equation of [1], in its original target coordinate T with branch values $0, \sqrt{-23}, -\sqrt{-23}$.

Theorem 7.1 (Explicit realization of the minimal-degree projection)**.** *There are explicit polynomials $J_0, J_1 \in \mathbf{Z}[T,V]$ and a primitive irreducible polynomial $P \in \mathbf{Z}[T,W]$ such that*

$$\deg_W P = 23, \qquad \deg_T P = 4,$$

and, in $\mathbf{Q}(X)$,

$$W = \frac{J_0(T,V)}{J_1(T,V)}, \qquad P(T,W) = 0, \qquad \mathbf{Q}(T,W) = \mathbf{Q}(T,V).$$

Consequently P defines the same regular M_{23}-cover. The projection $X \to \mathbf{P}^1_W$ has degree 4, and therefore explicitly realizes the minimum over $\mathbf{Q}$ established in [1, Remark 3.4].

This section constructs the earlier pencil $|K_X - A - B|$ explicitly. Unlike the osculating pencil, it is obtained by imposing simple vanishing at the two marked points. It gives a convenient minimal-degree plane equation and transports the Fano and affine residue fields exactly. The polynomial P and the sections J_0, J_1 in the theorem refer only to this pencil, not to the auxiliary S_4 map.

Set $D = T^2 + 23$. The coefficient of V^{23} in F is D^4, so

$$\Phi := F, \qquad \widehat{F} := D^{-4}F \in \mathbf{Q}(T)[V]$$

defines the same function field and is monic of degree 23 in V. We write C_0 for the affine plane curve $\Phi = 0$ and X for its normalization. The compactification under consideration is the closure in $\mathbf{P}^1_T \times \mathbf{P}^1_V$.

7.1. **The two points above $T^2 + 23 = 0$.** Let $b, c \in X(\overline{\mathbf{Q}})$ be the unique points over the roots of D. They are conjugate over $\mathbf{Q}$ and each has ramification index 23 for the T-map [1, Section 3].

Lemma 7.2 (Valuations at the two ramification points)**.** *At either $q \in \{b, c\}$ one has*

$$\operatorname{ord}_q(T - T(q)) = 23, \qquad \operatorname{ord}_q(D) = 23, \qquad \operatorname{ord}_q(V) = -4,$$

and

$$\operatorname{ord}_q\left(\frac{dT}{\Phi_V}\right) = 18.$$

Proof. The first two equalities follow from total ramification and the fact that D has a simple zero at $T(q)$. The coefficients of Φ, viewed as a polynomial in V, have a Newton-polygon edge joining $(0,0)$ and $(23,4)$ at $D = 0$. Hence its unique branch has $\operatorname{ord}_q(V) = -4$.

For completeness, the terms of Φ_V on this branch have orders

$$23 \operatorname{ord}_D(ia_i) - 4(i-1),$$

where $a_i(T)$ is the coefficient of V^i. The minimum is 4, attained uniquely by differentiating the D^4V^{23} term; all other orders are at least 5. Thus $\operatorname{ord}_q(\Phi_V) = 4$. Finally $\operatorname{ord}_q(dT) = 23 - 1 = 22$, so $\operatorname{ord}_q(dT/\Phi_V) = 22 - 4 = 18$. $\square$

The calculation is reproduced directly from the normalized coefficient table by `verification/verify_boundary_valuations.gp`.

For a polynomial $A(T, V)$ consider the adjoint differential

$$\omega_A = \frac{A(T,V)\,dT}{\Phi_V(T,V)}.$$

The interior of the bidegree-$(8, 23)$ rectangle gives the usual support $0 \le i \le 6$, $0 \le j \le 21$ for monomials T^iV^j. Regularity at b and c sharpens this $7 \cdot 22 = 154$-dimensional space substantially.

Lemma 7.3 (Adjoints regular at b and c)**.** *The differentials ω_A having interior support and regular at both b and c are represented by the 88-dimensional space*

$$\begin{aligned}\mathcal{A} = \bigoplus_{j=0}^{4} V^j\mathbf{Q}[T]_{\le 6} \;\oplus \bigoplus_{j=5}^{10} DV^j\mathbf{Q}[T]_{\le 4} \\ \oplus \bigoplus_{j=11}^{16} D^2V^j\mathbf{Q}[T]_{\le 2} \;\oplus \bigoplus_{j=17}^{21} D^3V^j\mathbf{Q}.\end{aligned}$$

In particular,

$$\dim \mathcal{A} = 35 + 30 + 18 + 5 = 88.$$

Proof. Write $A = \sum_{j=0}^{21} a_j(T)V^j$. By lemma 7.2, a term $D^m c(T)V^j\,dT/\Phi_V$ has order

$$23m - 4j + 18$$

at b and c, unless c also vanishes there. Regularity therefore requires

$$m \ge \left\lceil \frac{4j-18}{23} \right\rceil,$$

giving respectively $m = 0, 1, 2, 3$ on the four displayed blocks. Terms having distinct exponents j cannot cancel: their orders are distinct modulo 23. Divisibility by D^m lowers the residual T-degree from 6 to $6 - 2m$. Counting coefficients gives the stated dimension. □

7.2. **The affine nodes and the canonical system.** At an ordinary node, Rosenlicht's adjoint condition [7] is the vanishing of the numerator A. Thus each node contributes one linear evaluation condition, and the rank computation below verifies independence. We impose the conditions in one stroke using the reduced singular scheme modulo 31.

Exact computation 7.4 (The node graph modulo 31)**.** Over $\mathbf{F}_{31}$ the Jacobian ideal

$$(\Phi, \Phi_T, \Phi_V)$$

has lexicographic basis

$$h_{84}(T), \qquad V - s(T),$$

where $\deg h_{84} = 84$ and h_{84} is squarefree. The determinant of the Hessian is a unit on this scheme. Thus the affine singular scheme consists geometrically of 84 reduced ordinary nodes.

This statement is certified by `verification/verify_nodes_mod31.sing`; the explicit h_{84} and s are in `data/node_graph_mod31.inc`. Evaluation of the basis in lemma 7.3 on the node graph gives an 84×88 matrix over $\mathbf{F}_{31}$.

Exact computation 7.5 (Canonical kernel and degree-4 pencil)**.** The node-evaluation matrix has rank 84, so its kernel has dimension 4. This is the canonical adjoint space $H^0(X, K_X)$. On this kernel, vanishing at $b + c$ has rank 2, leaving the two-dimensional space $H^0(X, K_X - b - c)$ and hence a pencil.

Explanation of the two conditions at b and c. Among the blocks of lemma 7.3, the only term that can have order zero at b or c is

$$D^2(c_0 + c_1 T + c_2 T^2) V^{16}.$$

Its vanishing at both roots of D is equivalent to divisibility of the quadratic coefficient by D, namely

$$c_1 = 0, \qquad c_0 = 23 c_2.$$

These two conditions have rank 2 on the four-dimensional node kernel. The ranks are reproduced by `verification/verify_adjoint_mod31.py`. □

The successive dimensions are therefore

$$154 \longrightarrow 88 \longrightarrow 4 \longrightarrow 2.$$

The last two arrows are certified by full-rank minors modulo 31. This good-reduction calculation is the modular construction step; it does not by itself certify the characteristic-zero equation. That role is played by the exact function-field identity and irreducibility proof in section 7.4.

If ω_0 and ω_1 form a basis of $H^0(X, K_X - b - c)$, then their ratio $W = \omega_0/\omega_1$ is a rational function on X. The resulting row-reduced-echelon-form normalized pencil was reconstructed over $\mathbf{Q}$ by the Chinese remainder theorem. The least common multiple of the reconstructed coefficient denominators is 91; clearing it gives $J_0, J_1 \in \mathbf{Z}[T, V]$. Their degrees in V are 20 and 21, respectively, and each has degree 6 in T.

7.3. **Reconstruction of the bidegree $(23, 4)$ equation.** Set $W = J_0/J_1$. Eliminating V between $\Phi(T, V)$ and $J_0(T, V) - W J_1(T, V)$ means extracting the desired factor from

$$\mathrm{Res}_V\big(\Phi(T, V), J_0(T, V) - W J_1(T, V)\big).$$

We performed this calculation modulo good primes in a stable row-reduced-echelon-form (RREF) normalization, reconstructed the coefficients, and then discarded the numerical evidence in favor of the exact certificate below.

For transparency, the equation used a 575-bit CRT modulus. All 120 coefficients reconstructed and reduced correctly at the independent holdout prime 500015017. The pencil used a 604-bit modulus and passed independent holdout prime 137. These holdouts are evidence about reconstruction, not ingredients in the proof of theorem 7.1; their complete records are `data/f4_crt_reconstruction.json` and `data/pencil_crt_reconstruction.json`. As an auxiliary geometric check, the reduction of P modulo 31 divides the displayed defining resultant; this is verified by `verification/verify_f4_mod31.sing`.

7.4. **Exact certificate and the known degree minimum.** Write

$$P(T, W) = \sum_{j=0}^{23} p_j(T) W^j.$$

Proposition 7.6 (Exact function-field identity)**.** *In $\mathbf{Q}(T)[V]/(\Phi) = \mathbf{Q}(T)[V]/(\widehat{F})$ one has*

$$\sum_{j=0}^{23} p_j(T) J_0^j J_1^{23-j} = 0.$$

Equivalently, $P(T, J_0/J_1) = 0$ in $\mathbf{Q}(X)$.

Proof. This is a polynomial identity modulo $\Phi = F$, equivalently modulo the monic degree-23 relation $\widehat{F}$ over $\mathbf{Q}(T)$. The supplied Singular certificate evaluates the left side by homogeneous Horner reduction and obtains zero. The SageMath certificate independently rebuilds Φ, P, J_0, J_1 from JSON coefficient tables, reads no Singular inputs, and performs the same calculation in its own polynomial arithmetic. A self-contained Magma certificate repeats the quotient-ring identity from the same canonical tables. All computations are exact. □

Proposition 7.7 (Irreducibility)**.** *The polynomial P is primitive and irreducible in $\mathbf{Q}[T, W]$. In particular it is irreducible in $\mathbf{Q}(T)[W]$.*

Proof. The coefficient content is 1. Its reduction modulo 31 retains bidegree $(23, 4)$ and is irreducible in $\mathbf{F}_{31}[T, W]$, as checked independently by Singular, SageMath, and Magma. Gauss's lemma proves the claim. □

Proof of Theorem 7.1. By proposition 7.6, $\mathbf{Q}(T, W)$ is a subfield of $\mathbf{Q}(T, V)$. By proposition 7.7,

$$[\mathbf{Q}(T, W) : \mathbf{Q}(T)] = 23.$$

The source equation also gives $[\mathbf{Q}(T, V) : \mathbf{Q}(T)] = 23$, hence the inclusion is an equality. The regular M_{23} monodromy follows from [1, Proposition 3.5 and Corollary 3.6].

Regard P as a polynomial in T over $\mathbf{Q}(W)$. Since P is irreducible and has positive T-degree, Gauss's lemma and $\deg_T P = 4$ give $[\mathbf{Q}(T, W) : \mathbf{Q}(W)] = 4$. Thus W has degree 4 on X. That this degree is minimal over $\mathbf{Q}$ is the nonsplit-quadric argument already given in [1, Remark 3.4]: every map of degree at most 3 on the geometric genus-4 curve comes from a ruling of its canonical quadric, but neither ruling is defined over $\mathbf{Q}$. Our calculation in proposition 8.22 identifies the quadratic ruling field explicitly. □

8. Arithmetic of the distinguished branch fibre

The minimal equation and the original plane equation describe the same cover, but their coordinates need not display a singular branch fibre in the same way. We first transport the fibre over $T = 0$ exactly between the two models. We then determine its local monodromy, Fano-plane incidence, number fields, and Galois-selected flags, before asking how much of this internal arithmetic can explain the distinguished point of the seven-cover moduli problem.

8.1. **Transport to the minimal-degree coordinate.**

Theorem 8.1 (Transport of the fibre above $T = 0$)**.** *There are primitive irreducible polynomials $\widetilde{H}_7, \widetilde{R}_8 \in \mathbf{Z}[W]$ of degrees 7 and 8 such that*

$$F(0, V) = 23^4 H_7(V) R_8(V)^2,$$
$$P(0, W) = -23^3 \widetilde{H}_7(W) \widetilde{R}_8(W)^2.$$

Moreover

$$\gcd\big(J_1(0, V), H_7(V) R_8(V)\big) = 1,$$

and substitution $W = J_0(0, V)/J_1(0, V)$ induces isomorphisms

$$\mathbf{Q}[W]/(\widetilde{H}_7) \xrightarrow{\sim} \mathbf{Q}[V]/(H_7),$$
$$\mathbf{Q}[W]/(\widetilde{R}_8) \xrightarrow{\sim} \mathbf{Q}[V]/(R_8).$$

These isomorphisms give Galois-equivariant bijections between the respective sets of geometric roots.

The factors in the source coordinate are

$$\begin{aligned} H_7(V) &= V^7 - 312V^6 + 43848V^5 - 3914896V^4 + 241123008V^3 \\ &\qquad - 9876755712V^2 + 251799776256V - 2694891036672, \\ R_8(V) &= V^8 + 156V^7 - 1290V^6 - 1217744V^5 - 40611765V^4 \\ &\qquad + 2044635024V^3 + 115083889782V^2 + 968032514052V - 6058197515007. \end{aligned}$$

Both are irreducible over $\mathbf{Q}$. The corresponding primitive factors in the minimal-degree coordinate are

$$\begin{aligned} \widetilde{H}_7(W) = {} & 5259031199210877419615 5W^7 - 838025681952597547830 4W^6 \\ & + 5545714000161336126 72W^5 - 193386397022083274 24W^4 \\ & + 3678837111590735 36W^3 - 3404298821468160W^2 \\ & + 7203244769280W + 62710185984, \end{aligned}$$

and

$$\begin{aligned} \widetilde{R}_8(W) = {} & 1055142818078345829104 7W^8 - 2504505235463133310560W^7 \\ & + 258988485843116269632W^6 - 15238022132962377728W^5 \\ & + 557642314807148032W^4 - 12985270524477440W^3 \\ & + 187639106142208W^2 - 1535652528128W + 5439225856. \end{aligned}$$

Proof of Theorem 8.1. Reconstruct F, P, J_0, J_1 from their canonical integer coefficient tables. Exact factorization over $\mathbf{Q}$ gives the two displayed identities and proves the irreducibility of all four factors. Euclidean reduction gives

$$\gcd(J_1(0,V), H_7R_8) = 1,$$

so the specialized rational function is defined at every geometric point of the fibre.

Let v_7 be the image of V in $K_7 = \mathbf{Q}[V]/(H_7)$ and put

$$w_7 = J_0(0,v_7)/J_1(0,v_7).$$

Exact minimal-polynomial computation gives

$$\mathrm{minpoly}_{\mathbf{Q}}(w_7) = \frac{\widetilde{H}_7}{52590311992108774196155}.$$

In particular w_7 has degree 7 and generates K_7. Substitution therefore defines the first displayed field isomorphism. The identical calculation in $K_8 = \mathbf{Q}[V]/(R_8)$ gives

$$\mathrm{minpoly}_{\mathbf{Q}}\left(\frac{J_0(0,v_8)}{J_1(0,v_8)}\right) = \frac{\widetilde{R}_8}{10551428180783458291047},$$

and proves the second isomorphism. Applying all embeddings into $\overline{\mathbf{Q}}$ shows that the maps on roots are Galois-equivariant and bijective. Every step is reproduced by the certificate described in section 11. □

Remark 8.2. The factor degrees provide a consistency check. The theorem also requires regularity of the rational coordinate throughout the fibre and generation of the full residue fields. The coprimality and minimal-polynomial checks verify these two properties.

8.2. **Local monodromy at the branch point $T = 0$.**

8.2.1. *Group-theoretic notation.* Fix a copy of $M_{23} \subset \mathrm{Sym}(23)$ and elements
$$x \in 2A, \qquad y \in 23A, \qquad xy \in 23A,$$
so that $(x, y, (xy)^{-1})$ is a generating triple with class tuple $(2A, 23A, 23B)$. Set
$$K = O_2(C_{M_{23}}(x)), \qquad H = \mathrm{Fix}(x) \subset \{1, \ldots, 23\}.$$
Since x has cycle shape $1^7 2^8$, $|H| = 7$; the fixed letters form a heptad in the Witt design $S(4, 7, 23)$. One has
$$C_{M_{23}}(x) \cong 2^4{:}\,\mathrm{PSL}_3(2), \qquad |C_{M_{23}}(x)| = 2688, \qquad |K| = 16.$$
For reproducibility, the certificate labeling uses
$$\begin{aligned} x &= (3\,21)(4\,16)(9\,22)(10\,15)(11\,20)(12\,19)(13\,18)(14\,17), \\ y &= (1\,2\,11\,10\,16\,9\,6\,3\,23\,19\,20\,14\,21\,17\,4\,8\,22\,5\,18\,15\,13\,7\,12), \\ r &= (3\,18)(4\,16)(5\,6)(7\,23)(10\,14)(12\,19)(13\,21)(15\,17). \end{aligned}$$
The involution r will compare the displayed triple with its complex conjugate in theorem 8.20. These representatives generate the same Nielsen class as the tuple given in [1, Section 3]; all assertions below are invariant under simultaneous conjugation.

For fixed y let
$$X_y = \{x' \in 2A : x'y \in 23A\}, \qquad |X_y| = 161 = 7 \cdot 23,$$
whose $\langle y \rangle$-translation orbits are the seven Nielsen classes. We call the class containing x the *distinguished class*; it is Nielsen class 6 in the enumeration used below. This is an internal label for the class specified by the displayed permutation x. The exact Hurwitz algebra calculation in section 3 identifies it as the degree-one component over K_0; theorem 1.1 characterizes it by osculating incidence.

Put $K_0 = \mathbf{Q}(\sqrt{-23})$. Let $\mathcal{H}_{\mathrm{in}}$ denote the reduced zero-dimensional inner Hurwitz scheme over K_0 for normalized three-point covers with branch-cycle classes $(2A, 23A, 23B)$. Here *inner* means that generating triples are taken modulo simultaneous conjugation by M_{23}; we use the standard Hurwitz-space and Nielsen-class formalism of [6]. Its seven geometric points correspond to the seven $\langle y \rangle$-orbits above. In section 3 we compute a finite K_0-scheme from exact equations and then identify it with $\mathcal{H}_{\mathrm{in}}$ by certified branch cycles.

8.2.2. *Branch points and the fibre over $T = 0$.* The cover is branched over 0 and the two roots of $T^2 + 23$, with local monodromy $(2A, 23A, 23B)$ and total ramification at the order-23 points [1, Section 3]. The valuations and ordinary nodes were computed in lemma 7.2 and Exact computation 7.4. The discriminant separates the branch fibres from the projection singularities.

Proposition 8.3. *The integral model $F \in \mathbf{Z}[T, V]$ satisfies*
$$\mathrm{Res}_V(F, F_V) = uT^8(T^2+23)^{92}G(T)^2, \qquad \mathrm{disc}_V(F) = -uT^8(T^2+23)^{88}G(T)^2, \qquad \deg G = 84,$$
for some $u \in \mathbf{Q}^\times$, where G is squarefree and divides $\gcd(\mathrm{Res}_V(F, F_V), \mathrm{Res}_V(F, F_T))$, while T does not. Consequently:

1. *the fibre over $T = 0$ consists of smooth points of the plane model, and $F(0, V) = 23^4 H_7 R_8^2$ gives cycle type $1^7 2^8$, the shape of $2A$;*
2. *the zeros of G account for projection singularities rather than branch points of the normalization; their good reduction modulo 31 is the 84-node scheme of Exact computation 7.4;*
3. *at $T^2 + 23$ the Newton polygon is a single segment of slope $4/23$, so the local extension is totally ramified of degree 23.*

Riemann–Hurwitz gives
$$2g - 2 = -46 + 8 + 22 + 22 = 6, \qquad g = 4.$$

For comparison, direct factorization for the minimal-degree equation gives, up to a nonzero rational constant,

$$\operatorname{disc}_W P = T^8(T^2+23)^{22}H_{62}(T)^2, \qquad \deg H_{62} = 62.$$

The factor T^8 records genuine ramification in both models. The residual square factors and the exponent at T^2+23 depend on the projection.

8.2.3. *Why the cyclotomic obstruction vanishes.*

Proposition 8.4. *y^λ is conjugate to y in M_{23} if and only if λ is a quadratic residue modulo 23; for the eleven nonsquares it lies in the opposite class $23B$.*

The two order-23 branch points are conjugate over $\mathbf{Q}$ and individually rational over $\mathbf{Q}(\sqrt{-23})$; the quadratic subfield of $\mathbf{Q}(\zeta_{23})$ fixed by the squares. Thus the branch swap matches the nonsquare part of the cyclotomic character. Fried's branch-cycle argument sends (C_1, C_2, C_3) under σ to $(C_1^\lambda, C_2^\lambda, C_3^\lambda)$, with $\lambda = \chi(\sigma)$. For nonsquare λ the tuple becomes $(2A, 23B, 23A)$ while the same σ exchanges the two nonrational branch points. The classical obstruction therefore vanishes, but this permits field of moduli $\mathbf{Q}$ without forcing the fixed point of [1].

8.3. **The Fano plane on the seven unramified points.** The factors H_7 and R_8 were displayed in section 8.1. Their roots are respectively the seven unramified points and eight ramification points above $T=0$.

Theorem 8.5. $\operatorname{Gal}(H_7) \cong \operatorname{PSL}_3(2)$, *acting on the seven unramified sheets. The* 21 *involutions each fix exactly three sheets. There are seven distinct fixed triples, each arising from three involutions, and these triples form a* 2-$(7,3,1)$ *design. Thus the sheets are the points of a Fano plane and the fixed triples are its lines.*

A *flag* means an incident point–line pair in this plane.

Proposition 8.6. *Let $\alpha_1, \dots, \alpha_7$ be the roots of H_7. The three-subset-sum resolvent*

$$\mathcal{R}_3(Y) = \prod_{1\le i<j<k\le 7} \bigl(Y - (\alpha_i+\alpha_j+\alpha_k)\bigr)$$

factors over $\mathbf{Q}$ into irreducible factors of degrees 7 *and* 28. *The degree-seven factor is*

$$\begin{aligned} L_7(Y) = Y^7 &- 936Y^6 + 379728Y^5 - 86588640Y^4 + 11922182400Y^3 \\ &- 982633068288Y^2 + 44572755885824Y - 859311381083136. \end{aligned}$$

Its roots are the sums along the seven Fano lines. The fields $\mathbf{Q}[V]/(H_7)$ and $\mathbf{Q}[Y]/(L_7)$ are nonisomorphic, but they have the same Galois closure and Dedekind zeta function. Their common field discriminant is

$$2^4 3^6 23^6 = 1726690609296.$$

Their point and line stabilizers are nonconjugate subgroups of order 24 *with equal permutation characters, the classical Gassmann pair for* $\operatorname{PSL}_3(2)$*; the resulting equality of Dedekind zeta functions is the Gassmann–Perlis criterion* [8].

Corollary 8.7. *The roots of $\widetilde{H}_7$ carry the transported Fano-plane point structure. Its line field is defined by L_7, and the point and line fields remain nonisomorphic arithmetically equivalent septics.*

Proof. The first isomorphism of theorem 8.1 identifies the two seven-element Galois sets. Fixed triples and incidence are functorial under this identification. □

Proposition 8.8. *The points, lines, and flags admit the following description through $C_{M_{23}}(x)$. All 99 involutions of $C_{M_{23}}(x)$ belong to class $2A$, and:*

1. *the fifteen nonidentity elements of K fix H pointwise. The central element x is one conjugacy class and the other fourteen form a second. Writing $U = K/\langle x\rangle \cong \mathbf{F}_2^3$, the nonzero elements of U identify with the seven lines; the plane is $\mathbf{P}(U^*)$ on points and $\mathbf{P}(U)$ on lines;*
2. *as an $\mathbf{F}_2[C_{M_{23}}(x)]$-module, K is indecomposable, with unique minimal nonzero submodule $\langle x\rangle$. Each nonzero element of U has two preimages in K, exchanged by multiplication by x;*
3. *the 84 involutions of $C_{M_{23}}(x) \smallsetminus K$ form one class and act on H as transvections. The map to center and fixed line is a surjection onto the 21 flags, with four involutions above each flag. For such t,*

$$C_K(t) = \langle x\rangle \cup \{\textit{the two lifts of the axis}\}, \qquad \mathrm{Fix}(t) \cap H = \mathrm{axis}(t).$$

There is no $C_{M_{23}}(x)$-invariant complement to $\langle x\rangle$ in K. Moreover the outer point–line duality of $\mathrm{PSL}_3(2)$ is not induced by an automorphism of $C_{M_{23}}(x)$ or by a permutation of the 23 sheets normalizing $C_{M_{23}}(x)$:

$$\mathrm{Aut}(C_{M_{23}}(x)) = \mathrm{Inn}(C_{M_{23}}(x)) \quad (\text{order } 1344), \qquad N_{\mathrm{Sym}(23)}(C_{M_{23}}(x)) = C_{M_{23}}(x).$$

Thus no such symmetry exchanges the point and line Galois sets.

8.4. **Number fields obtained from the fibre.** Put

$$E = \mathrm{Spl}_{\mathbf{Q}}(H_7), \qquad L_0 = \mathrm{Spl}_{\mathbf{Q}}(H_7R_8).$$

We construct a third field M and obtain

$$\mathbf{Q} \subset E \subset L_0 \subset M, \qquad [E : \mathbf{Q}] = 168, \quad [L_0 : E] = 8, \quad [M : L_0] = 2.$$

The successive groups over $\mathbf{Q}$ are $\mathrm{PSL}_3(2)$, $2^3{:}\,\mathrm{PSL}_3(2)$, and $2^4{:}\,\mathrm{PSL}_3(2)$. The corresponding permutation actions are, respectively, those on the seven unramified points, on the eight ramification points, and on the two local branches at each ramification point.

8.4.1. *The splitting field of H_7R_8.* The local decomposition group centralizes x. Since x exchanges the two local branches at a ramification point but fixes its root of R_8, the action on the eight roots factors through $C_{M_{23}}(x)/\langle x\rangle$. The translation subgroup is $U = K/\langle x\rangle \cong \mathbf{F}_2^3$. After choosing an affine origin, the difference of two roots is a nonzero vector of U, hence a Fano line by proposition 8.8; each of the seven nonzero differences occurs for four of the 28 unordered pairs.

Theorem 8.9. *The splitting field of H_7R_8 has Galois group $2^3{:}\,\mathrm{PSL}_3(2)$ of order 1344, acting affinely on the eight roots of R_8. The restriction $\mathrm{Gal}(L_0/\mathbf{Q}) \to \mathrm{Gal}(E/\mathbf{Q})$ has kernel $K/\langle x\rangle \cong (\mathbf{Z}/2)^3$.*

Proof. Decomposition theory at the tame rational place, together with equality of arithmetic and geometric monodromy [1], embeds $\mathrm{Gal}(L_0/\mathbf{Q})$ in $C_{M_{23}}(x)/\langle x\rangle$. Restriction to E is surjective, and its kernel is a submodule of the irreducible three-dimensional module U, hence either trivial or all of U. At $p = 599$, H_7 and L_7 split completely while R_8 factors into four quadratics. Frobenius is therefore trivial on E but a nonzero translation on the roots of R_8, proving that the kernel is U. ∎

Independently, a direct number-field computation gives $[\mathrm{Spl}_{\mathbf{Q}}(R_8) : \mathbf{Q}] = 1344$.

8.4.2. *Affine frames and specialization.* Let A be the eight roots of R_8. An ordered affine frame is a tuple (a_0, a_1, a_2, a_3) whose three differences $a_i - a_0$ form a basis of U. Let

$$\mathcal{F}(A) = \{\text{ordered affine frames of } A\}/U.$$

Proposition 8.10. *$\mathcal{F}(A)$ has 168 elements. The action of $C_{M_{23}}(x)/\langle x\rangle$ descends to a free transitive action of $C_{M_{23}}(x)/K \cong \mathrm{PSL}_3(2)$. Via proposition 8.8, a translation class of frames is equivalently an ordered basis of the Fano line module.*

Proof. There are $8 \cdot 7 \cdot 6 \cdot 4 = 1344$ affine frames. Translation acts freely on the eight origins, leaving 168. The quotient $\mathrm{PSL}_3(2) = \mathrm{GL}_3(2)$ acts freely and transitively on ordered bases of U. □

Let S be the marked geometric generic fibre, based at the tangential basepoint $T \to 0$. For $\sigma \in \mathfrak{G}_{\mathbf{Q}}$, let $d(\sigma)$ be its descent permutation. Since the inertia x has order two, the branch-cycle lemma places $d(\sigma)$ in $C_{M_{23}}(x)$. Write $\rho_8 : \mathfrak{G}_{\mathbf{Q}} \to \mathrm{Sym}(A)$ for the Galois action on the roots of R_8.

Proposition 8.11 (Tame specialization at $T = 0$)**.** *For every $\sigma \in \mathfrak{G}_{\mathbf{Q}}$,*

$$\rho_8(\sigma) = \mathrm{pair}(d(\sigma)),$$

where pair *is the induced action on the eight two-sheet inertia orbits. Consequently the induced action on $\mathcal{F}(A)$ is the one obtained from the generic-fibre descent permutation.*

Proof. Let $R = \mathbf{Q}[T]^h_{(T)}$ and normalize $\operatorname{Spec} R$ in the cover. At a geometric point of ramification index e, tameness and smoothness give the completed local form $T = u^e$ after extracting an eth root of a unit. The e sheets form one inertia orbit and specialize to one point. Hence specialization gives a canonical Galois-equivariant bijection

$$S/\langle x\rangle \xrightarrow{\sim} \{\text{geometric points above } T = 0\}.$$

Restricting to the eight two-element orbits proves the formula. This is the finite-set case of tame specialization [25, Exposé XIII, Section 2.10]. □

8.4.3. *Separating the two branches.* The group $C_{M_{23}}(x)$ is perfect, its center is $\langle x\rangle$, and the central extension

$$1 \longrightarrow \langle x\rangle \longrightarrow C_{M_{23}}(x) \longrightarrow C_{M_{23}}(x)/\langle x\rangle \longrightarrow 1$$

does not split. It acts faithfully and transitively on the sixteen local branches above the eight roots of R_8, with branch stabilizer isomorphic to $\mathrm{PSL}_3(2)$.

Near a root v of R_8, the two branches have expansions

$$V = v \pm \sqrt{e(v)T} + O(T), \qquad e(v) = -\frac{2F_T(0,v)}{F_{VV}(0,v)}.$$

Since $F_{VV}(0,v) = 2 \cdot 23^4 H_7(v) R_8'(v)^2$, the square class of $e(v)$ is represented by

$$c(V) \equiv -F_T(0,V)H_7(V) \pmod{R_8}.$$

Theorem 8.12. *The eight values $c(v)$ represent one nontrivial square class in $L_0^\times/L_0^{\times 2}$. If $M = L_0(\sqrt{c(v)})$ for any root v, then*

$$\mathrm{Gal}(M/\mathbf{Q}) \cong C_{M_{23}}(x) = 2^4{:}\,\mathrm{PSL}_3(2), \qquad [M : L_0] = 2,$$

and the nontrivial element of $\mathrm{Gal}(M/L_0)$ *is x.*

Proof. Tame local theory identifies the splitting field over $\mathbf{Q}((T))$ as $\mathcal{L} = L_0((\sqrt{cT}))$, with Galois group $C_{M_{23}}(x)$. Since $C_{M_{23}}(x)$ is perfect, $\sqrt{T} \notin \mathcal{L}$ and c is not a square in L_0. After adjoining $\sqrt{T}$ the Galois group is $C_{M_{23}}(x) \times \mathbf{Z}/2$. The diagonal involution $(x, -1)$ fixes $\sqrt{c} = \sqrt{cT}/\sqrt{T}$, so taking constant fields gives

$$\mathrm{Gal}(M/\mathbf{Q}) \cong (C_{M_{23}}(x) \times \mathbf{Z}/2)/\langle (x, -1) \rangle \cong C_{M_{23}}(x).$$

Exact arithmetic checks additionally show that $X^2 - dc(V)$ is irreducible over $\mathbf{Q}[V]/(R_8)$ for

$$d \in \{\pm 1, \pm 2, \pm 3, \pm 6, \pm 23, \pm 46, \pm 69, \pm 138\}.$$

At each of fourteen primes below $3 \cdot 10^5$ splitting completely in L_0, the eight values have one quadratic character; at six, beginning with 13049, it is -1, independently certifying nonsquareness. $\square$

The polynomial R_8 has four real roots. Exactly two have $c(v) > 0$, so both local branches above them are real; the other two have $c(v) < 0$, so their branches are exchanged by complex conjugation. This agrees with the permutation r, which fixes the branches above two real ramification points and exchanges the branches above the other two.

Corollary 8.13. *Replacing (H_7, R_8) by $(\widetilde{H}_7, \widetilde{R}_8)$ leaves the fields E, L_0, M, the affine torsor, and the centralizer action unchanged under the transport induced by $W = J_0/J_1$.*

Proof. Theorem 8.1 identifies both residue Galois sets. The first two splitting fields and their functorial actions agree. The final quadratic extension belongs to the normalized cover and separates its local branches, independently of the coordinate labeling the residue points. $\square$

8.4.4. *Ramification.*

Theorem 8.14. *Let $E_8 = \mathbf{Q}[V]/(R_8)$ and $E_8' = E_8(\sqrt{c(v)})$. The fields $\mathbf{Q}[V]/(H_7)$, $\mathbf{Q}[Y]/(L_7)$, and E_8 have discriminants $2^4 3^6 23^6$, $2^4 3^6 23^6$, and $2^{16} 3^6 23^6$. The relative discriminant of E_8'/E_8 has norm $2^8 3^2 23^2$. In particular, M is unramified outside $\{2, 3, 23\}$. The complete lists of pairs (e, f) in the constituent fields are*

p	*either septic field*	E_8
23	$(7,1)$	$(1,1), (7,1)$
3	$(1,1), (3,1), (3,1)$	$(1,1), (1,1), (3,1), (3,1)$
2	$(1,1), (3,1), (3,1)$	$(4,2)$.

Thus the ramified septic primes at 2 are tame; the ramified primes at 3 and the octic prime at 2 are wild. At 23, inertia and decomposition in $L_0/\mathbf{Q}$ are C_7 and 7:3, respectively; in $M/\mathbf{Q}$ they are C_{14} and $C_2 \times (7{:}3)$.

Proof. The discriminants and prime decompositions are computed from exact maximal orders. Since M is the normal closure obtained by adjoining the conjugates of the local square root defining E_8'/E_8, their discriminant support gives the asserted bound for M.

At 23 the constituent extensions are tame. The Galois closure is therefore tame, with cyclic inertia acting on the octic roots as one 7-cycle and one fixed point; its inertia is C_7. The normalizer of this subgroup in $2^3{:}\mathrm{PSL}_3(2)$ is 7:3. Tame local theory gives $\phi\tau\phi^{-1} = \tau^{23} = \tau^2$; the exponent 2 has order three modulo 7, so the decomposition group is the full 7:3.

The relative discriminant of E_8'/E_8 has exponent one at each of the two primes above 23. Those quadratic extensions are ramified. The ramification index of L_0/E_8 at either chosen prolongation is either 7 or 1, hence odd, so the quadratic ramification persists. Thus inertia in M has order 14 and is cyclic by tameness. The decomposition group is a central extension of 7:3 by C_2. Since 21 is odd this extension splits, giving $C_2 \times (7{:}3)$. $\square$

The relative discriminant proves ramification of E_8'/E_8 at 2 and 3, but does not by itself prove its persistence after base change to L_0. We make no assertion here about the full inertia groups in M at those two primes. In particular, a census restricted to order-12 subgroups cannot establish that the actual inertia at 2 has order 12.

8.5. **Flags selected by complex conjugation and Frobenius.**

Theorem 8.15. *Every involution of* $\mathrm{PSL}_3(2)$ *acts as a transvection. It fixes one line pointwise, its axis, and has a distinguished center on that line. The map from involutions to center–axis pairs is a bijection onto the* 21 *flags.*

On points the cycle type is $1^3 2^2$. *Besides the axis, exactly two lines are invariant, each consisting of the center and one transposed pair. The group is transitive on flags; both a flag stabilizer and an involution centralizer have order* 8.

The field E *has no real embeddings and* 84 *conjugate pairs of complex embeddings. Each complex place determines the flag of its conjugation involution, giving a four-to-one map from the* 84 *places to the* 21 *flags. The same construction applies at an unramified finite place whose Frobenius is an involution.*

Corollary 8.16. *For complex conjugation, the axis is the set of three real roots of* H_7. *Their sum is a root of* L_7. *The center is the unique real root lying on the other two invariant lines; numerically it is the largest real root, approximately* 102.2684.

If p is unramified and H_7 has factorization type $1^3 2^2$ modulo p, then Frobenius is an involution. Such primes have Chebotarev density 1/8.

Theorem 8.17. *Let* N_{28} *be the degree-28 nonline factor of* $\mathcal{R}_3$, *and exclude primes dividing* $\mathrm{Res}(L_7, N_{28})$. *At every remaining unramified prime where* H_7 *has type* $1^3 2^2$, *the three roots in* $\mathbf{F}_p$ *sum to a root of* L_7 *and form the axis. The other four roots form two conjugate pairs. Counting, for each fixed root, how many pairs complete a Fano line gives* $(0, 0, 2)$; *the root with count* 2 *is the center. Exact computation checks all* 9,889 *such primes below* 10^6. *Below that bound the only excluded collision is* $p = 22159$.

The real axis is defined over $\mathbb{R}$ but not over $\mathbf{Q}$, since $\mathrm{Gal}(E/\mathbf{Q})$ is transitive on the seven lines. At 23 one sees the complementary Singer-cycle picture: after cyclically labeling the seven points, one Fano line is a difference set and the other six are its translates.

8.6. **A first obstruction: geometric reflection.** The branch divisor is preserved by the rational base involution

$$\tau : \mathbf{P}^1_T \longrightarrow \mathbf{P}^1_T, \qquad T \longmapsto -T,$$

which fixes the $2A$ branch point and exchanges the two order-23 branch points. It does not, however, lift to any of the seven covers. The two endpoint inertia generators lie in the distinct classes $23A$ and $23B$. A holomorphic branch swap preserves the orientation of a small loop and therefore preserves its inertia conjugacy class; inversion occurs only for an orientation-reversing comparison such as complex conjugation.

Proposition 8.18 (Geometric reflection obstruction)**.** *The holomorphic involution* $\tau : T \mapsto -T$ *has no lift to a cover in the Nielsen class* $(2A, 23A, 23B)$. *In contrast, the orientation-reversing comparison equations*

$$r \in C_{M_{23}}(x), \qquad y^r = z^{-1}, \qquad z^r = y^{-1}$$

have solution counts $(0, 0, 1, 0, 0, 1, 1)$ *on the seven normalized triples. Thus IDs* 3, 6, 7 *are the classes fixed by the chosen complex-conjugation action.*

Proof. For the natural degree-23 action, exact computation gives

$$C_{S_{23}}(M_{23}) = 1, \qquad N_{S_{23}}(M_{23}) = M_{23}.$$

For every normalized triple, y is not conjugate to z in M_{23}, whereas it is conjugate to z^{-1}. Any sheet permutation induced by a holomorphic lift would normalize M_{23} and carry a positive local inertia generator to a conjugate positive generator at the other endpoint. The normalizer identity would place it in M_{23}, contradicting the first class calculation. Complex conjugation reverses the local loop, so its finite comparison uses z^{-1} instead. Exhausting the displayed orientation-reversing equations gives the asserted counts. □

Let

$$Y_x = \{y'' \in 23A : xy'' \in 23A\}.$$

Pairs (x, y'') represent the seven Nielsen classes, and $C_{M_{23}}(x)$ acts on Y_x by conjugation.

Proposition 8.19. *The group M_{23} acts freely on the admissible pairs (x', y') with $x' \in 2A$, $y' \in 23A$, and $x'y' \in 23A$. Moreover*

$$|Y_x| = 18816 = 7 \cdot 2688,$$

and Y_x is the disjoint union of seven free $C_{M_{23}}(x)$-orbits, one per Nielsen class. Equivalently,

$$|Y_x| = |X_y| \frac{|23A|}{|2A|} = 161 \cdot \frac{443520}{3795} = 18816.$$

In these coordinates complex conjugation is

$$\mu : Y_x \longrightarrow Y_x, \qquad \mu(y'') = xy''.$$

Theorem 8.20. *The equation*

$$y''^w = xy'', \qquad w \in C_{M_{23}}(x),$$

has a solution exactly when the corresponding Nielsen class is fixed by complex conjugation. Exactly three classes are fixed. For each, the comparison element w is unique, lies in $C_{M_{23}}(x) \smallsetminus K$, and acts as a transvection on the roots of H_7. For the distinguished class, the unique solution is $w = r$; it satisfies $x^r = x$ and $y^r = xy$. Its center and axis give the real flag of theorem 8.15.

At the chosen real place, complex conjugation exchanges the same two branch points and induces this orientation-reversing permutation on the Nielsen set. The element r acts on the roots of H_7 with type $1^3 2^2$, fixes three Fano lines, and fixes four roots of R_8. Correspondingly, H_7, L_7, R_8 have $3, 3, 4$ real roots. The three fixed classes are the fixed set of this one complex-conjugation element. A $\mathbf{Q}$-defined cubic subscheme would require stability under the full absolute Galois group. The calculation neither establishes such stability nor explains why the distinguished class is globally fixed.

To relate the fibre arithmetic to the fixed-class problem, we now separate what is supplied by the unpointed branch-fibre structures from the additional relative geometric data that a specialization proof would require.

8.7. **Why unpointed branch-fibre data do not distinguish the fixed point.** The construction of [1] starts from a seven-element Nielsen class. Its Galois-fixed point produces the rational cover studied here. The Fano plane is attached instead to the seven unramified points in the chosen $2A$ branch fibre of that cover. Thus the results above start from the known rational cover and describe arithmetic internal to it; its descent is an input from [1].

The roles of the structures above may be summarized as follows.

(1) The minimal function W transports the residue fields between the two plane models, while the Fano incidence is intrinsic to the chosen fibre.

(2) The Fano plane is intrinsic to the seven fixed sheets of the chosen $2A$ fibre, but no individual Fano line is Galois invariant over $\mathbf{Q}$.
(3) The affine-frame torsor is compatible with tame specialization from the geometric generic fibre to the fibre over $T = 0$. Distinguishing the rational Nielsen class from the other six requires additional pointed descent data.

The coordinate comparison preserves all Fano and affine residue data, so the finite incidence calculations of §6.3 are unchanged by passing to the minimal-degree model. The branch fibre by itself still does not distinguish the degree-one Hurwitz point. A compatible construction over the local Hurwitz model, retaining the specified integral multiplicities, is not supplied by this coordinate change.

There is also an exact, entirely group-theoretic limit on what unpointed characteristic-23 inertia can determine. Let $I \leq M_{23}$ have order 23 and put $D = N_{M_{23}}(I)$. Direct computation in the supplied permutation model gives $D \cong 23{:}11$. The conjugates of the unpointed inertia subgroup I are therefore indexed by the 40320 points of M_{23}/D. Once a chosen local object is reduced to its unpointed order-23 inertia subgroup, its possible placements are exactly the points of this homogeneous space.

Proposition 8.21 (Unpointed 23-inertia obstruction)**.** *For each of the three complex-conjugation-fixed Nielsen classes, let r be the orientation-reversing comparison of proposition 8.18 and let*

$$A = C_{M_{23}}(x) \cap C_{M_{23}}(r).$$

Then $|A| = 32$, the three ordered pairs (x, r) are simultaneously conjugate in M_{23}, and A acts freely on the 40320 *conjugates of I, indexed by M_{23}/D. The involution r divides them into* 20160 *unordered pairs, on which $A/\langle r \rangle$ acts freely. Consequently there are* 1260 *orbits of reflection-pairs under $A/\langle r \rangle$ for each fixed Nielsen class. In particular, unpointed order-*23 *inertia together with (x, r) does not canonically select one pair or distinguish the distinguished class from the other two complex-conjugation-fixed classes.*

Proof. Exact computation gives $|A| = 32$ and one simultaneous-conjugacy orbit for the three anchors. Since $|D| = 253$ is odd, the stabilizer in the 2-group A of any coset in M_{23}/D is an intersection with a conjugate of D and is therefore trivial. Thus A acts freely. The element r is central in A and has no fixed coset. An element of A stabilizing one of its unordered two-element orbits either fixes a coset, hence is the identity, or sends it to its r-translate, hence is r. The induced $A/\langle r \rangle$-action is free, and

$$\frac{20160}{|A|/2} = \frac{20160}{16} = 1260.$$

$\square$

Thus a 23-adic argument must augment the two inertia subgroups with local–global comparison data. More precisely, the characteristic-23 Kummer coordinate has two C_{11}-orbits on its nonzero parameters: the squares in $\mathbf{F}_{23}^{\times}$ and their nonsquare coset. On the Fano side, if $F_{\mathrm{fl}} = E^{D_8}$ is a flag field, the two Klein four subgroups $V_4 \subset D_8$ give two quadratic extensions E^{V_4}/F_{fl}. A proposed explanation along these lines would need a Galois-equivariant comparison between these two explicitly defined two-element structures.

There is another natural quadratic datum on the curve, but it does not supply this comparison. Let (A_0, A_1, A_2, A_3) be the RREF-normalized basis of the canonical adjoint space reconstructed from the same 88 coefficient positions used above. Order quadratic monomials as

$$X_0^2, X_0X_1, X_0X_2, X_0X_3, X_1^2, X_1X_2, X_1X_3, X_2^2, X_2X_3, X_3^2.$$

The nonsquare discriminant of this canonical quadric is the obstruction used in [1, Remark 3.4] to prove that degree 4 is minimal over $\mathbf{Q}$. The next proposition refines that argument by identifying the ruling field exactly; that field is also relevant to the local–global comparison pursued here.

Proposition 8.22 (Canonical ruling field)**.** *The unique quadric containing the canonical image, normalized to have coefficient* 1 *at* X_3^2*, has coefficient vector*

$$\frac{1}{297}\left(\begin{matrix}442260599149123, -18897532752913, -34200754149, 794833324,\\ 168493762065, 745707549, -14145827, 660893, -31345, 297\end{matrix}\right).$$

If B is twice its symmetric Gram matrix, then

$$\det B = \frac{6444541387164161510278888}{970299} = 4873\left(\frac{38141181796}{3267}\right)^2, \qquad 4873 = 11 \cdot 443.$$

Consequently the two rulings are governed by

$$\mathbf{Q}(\sqrt{4873}),$$

*which is different from $K_0 = \mathbf{Q}(\sqrt{-23})$, the field over which the two order-*23 *branch points are ordered.*

Proof. The four canonical RREF vectors are reconstructed over $\mathbf{Q}$ from 39 good primes with a 604-bit CRT modulus; the ten quadric coefficients use a 141-bit modulus. Both reconstructions agree coefficient by coefficient at the independent holdout prime 137, and the canonical basis reduces to the directly recomputed basis modulo 31. Exact reduction in $\mathbf{Q}(T)[V]/(\widehat{F})$ annihilates the displayed quadratic combination of the ten products A_iA_j; their coefficient matrix has rank 9, proving uniqueness. Exact determinant evaluation gives the displayed square class. The components of the Fano scheme of lines on a smooth quadric surface are defined over its discriminant extension, so the ruling field is $\mathbf{Q}(\sqrt{4873})$. The SageMath and Magma certificates independently repeat the quotient-ring, rank, and determinant calculations. $\square$

The quadratic extension governing the two rulings cannot supply this comparison. Indeed, retain the notation $F_{\mathrm{fl}} = E^{D_8}$ and let E^{V_4}/F_{fl} be either of the two quadratic flag extensions just defined. If its square class were the product of the square classes obtained from $K_0/\mathbf{Q}$ and from the ruling extension $\mathbf{Q}(\sqrt{4873})/\mathbf{Q}$, then

$$E^{V_4} = F_{\mathrm{fl}}\mathbf{Q}(\sqrt{-23 \cdot 4873}).$$

The right side would imply $\mathbf{Q}(\sqrt{-23 \cdot 4873}) \subset E$. This is impossible because $\mathrm{Gal}(E/\mathbf{Q}) \cong \mathrm{PSL}_3(2)$ has no quotient of order two. Thus the ruling field does not provide the required local–global comparison.

The reflection, inertia, and ruling-field calculations determine the reach and the limits of the arithmetic internal to the distinguished cover. They do not by themselves construct the degree-one component of the seven-cover moduli problem. The local criterion of section 6.2 uses the normalized integral Hurwitz algebra; it is not a consequence of these unpointed branch-fibre data.

We now leave the branch fibre for an independent global application of the minimal equation.

9. Arithmetic specializations

Theorem 9.1 (Uniform specialization)**.** *For every integer*

$$t \equiv 2 \pmod{319},$$

the primitive part of $P(t, W)$ has degree 23*, and its splitting field over $\mathbf{Q}$ has Galois group M_{23} in its natural degree-*23 *action.*

At $T = 0$ the specialized polynomial is necessarily reducible and non-squarefree: it records the decomposition and inertia groups at a branch point, not the generic monodromy. Away from the branch locus the same degree-23 equation returns to the full Mathieu group.

We first record the specialization fact that identifies the ambient group.

Lemma 9.2 (Specialization subgroup)**.** *Let $Q(T, x) \in \mathbf{Q}[T, x]$ have splitting field $L/\mathbf{Q}(T)$ with Galois group G. If the leading coefficient and discriminant in x are nonzero at $t \in \mathbf{Q}$, then the splitting field of $Q(t, x)$ has Galois group isomorphic, in its action on the roots, to a subgroup of G.*

Proof. Work over $R = \mathbf{Q}[T]_{(T-t)}$. After dividing by its unit leading coefficient, Q is monic with unit discriminant. The normalization in L is unramified above $T = t$. The residue extension at a chosen prime has Galois group its decomposition group, a subgroup of G. The distinct specialized roots generate that residue extension, proving the claim. □

For an integer $t \equiv 2 \pmod{319}$, let c_t be the positive gcd of the coefficients of $P(t, x)$ and set

$$f_t(x) = c_t^{-1} P(t, x) \in \mathbf{Z}[x].$$

Proof of Theorem 9.1. Write $t = 2 + 319n$. Since $319 = 11 \cdot 29$,

$$P(t, x) \equiv P(2, x) \pmod{11}, \qquad P(t, x) \equiv P(2, x) \pmod{29}.$$

The leading coefficient is 5 modulo 11 and 22 modulo 29, so neither prime divides c_t and f_t retains degree 23.

Modulo 29, the monic coefficient vector in increasing order is

$$(5, 3, 7, 15, 21, 21, 17, 9, 1, 28, 21, 25, 3, 14, 20, 28,$$
$$5, 27, 24, 16, 9, 20, 18, 1).$$

Exact powering gives

$$x^{29^{23}} \equiv x \pmod{\bar{f}_{t,29}}, \qquad \gcd(\bar{f}_{t,29}, x^{29} - x) = 1.$$

Rabin's criterion proves irreducibility because 23 is prime. Thus f_t is irreducible over $\mathbf{Q}$, and Frobenius supplies a 23-cycle.

Modulo 11 one has

$$\bar{f}_{t,11} = g_2 g_7 g_{14},$$

where

$$\begin{aligned} g_2 &= x^2 + 8x + 3, \\ g_7 &= x^7 + 6x^6 + 7x^5 + 6x^4 + 8x^2 + 3x + 1, \\ g_{14} &= x^{14} + 8x^{13} + 6x^{12} + 8x^{11} + 6x^{10} + 6x^9 + 3x^8 + 2x^7 \\ &\quad + 10x^6 + 3x^5 + x^4 + 8x^3 + 10x^2 + 7x + 6. \end{aligned}$$

The factors are distinct and irreducible, so Frobenius supplies an element of cycle type $(2, 7, 14)$ and order 14. Squarefreeness also proves the characteristic-zero discriminant is nonzero, and lemma 9.2 and theorem 7.1 embeds $H_t = \mathrm{Gal}(f_t/\mathbf{Q})$ in M_{23}.

The ATLAS maximal-subgroup orders are [12, 13]

$$443520,\ 40320,\ 40320,\ 20160,\ 7920,\ 5760,\ 253.$$

Only $253 = 23 \cdot 11$ is divisible by 23, and it is odd. A subgroup containing both a 23-cycle and an element of order 14 is therefore not proper. Hence $H_t = M_{23}$. □

Corollary 9.3. *The explicit primitive polynomial $f_2(x) = P(2, x)$ has Galois group M_{23} over $\mathbf{Q}$.*

9.1. **Controlled ramification inside the progression.** Call a family mutually independent if every finite subfamily is linearly disjoint over $\mathbf{Q}$.

Huang et al. already deduce the abstract existence of an infinite mutually independent family from their regular realization [1, Corollary 1.4(d)]. The refinement below chooses such a family inside the explicit progression of theorem 9.1 while forcing a new, controlled inertia prime at each step.

Theorem 9.4. *Among the splitting fields of f_t with $t \equiv 2 \pmod{319}$ there is an infinite mutually independent collection of M_{23}-extensions. More precisely, outside a finite set of primes, if $q \nmid 319$ and*

$$t \equiv 2 \pmod{319}, \qquad v_q(t) = 1,$$

then inertia at q is generated by an element of class $2A$.

Proof. Beckmann's specialization inertia theorem [14, Proposition 4.2], applied at $T = 0$, says that away from finitely many primes, meeting the branch point with multiplicity m produces inertia conjugate to $\langle g^m \rangle$ for local monodromy g. Enlarge this exceptional set to contain 23. If $v_q(t) = 1$, then $T^2 + 23$ is a unit at the specialization, and the inertia group is the class-$2A$ group $\langle g \rangle$.

The Chinese remainder theorem gives

$$t \equiv 2 \pmod{319}, \qquad t \equiv q \pmod{q^2}.$$

Choose each new q outside the bad set and the ramified primes of all previous fields. The new field ramifies at q while the preceding fields do not, so the fields are pairwise distinct.

For mutual independence, suppose $L_1, \ldots, L_r$ are linearly disjoint and $K_r = L_1 \cdots L_r$, with Galois group M_{23}^r. For a new field L, $K_r \cap L$ is Galois over $\mathbf{Q}$. Simplicity of M_{23} makes it either $\mathbf{Q}$ or L. In the latter case a surjection $M_{23}^r \twoheadrightarrow M_{23}$ must factor through one coordinate: the coordinate images are commuting normal subgroups, and simplicity with trivial center permits only one nontrivial image. Then L equals a previous L_i, contradicting distinctness. Hence $K_r \cap L = \mathbf{Q}$. ☐

Exact computation 9.5 (A second certified field)**.** The integer

$$3830 \equiv 2 \pmod{319}, \qquad 3830 \equiv 5 \pmod{25}$$

lies in the progression and meets $T = 0$ 5-adically with multiplicity one. Over $\mathbf{F}_5$, the repeated part

$$\gcd(F(0, V), F_V(0, V))$$

has degree 8, is squarefree, and is coprime to $F_T(0, V)$. Exact maximal-order computations give

$$v_5(\mathrm{Disc}(\mathbf{Q}[x]/(f_2))) = 0, \qquad v_5(\mathrm{Disc}(\mathbf{Q}[x]/(f_{3830}))) = 8.$$

Thus the $t = 3830$ splitting field ramifies at 5 while the $t = 2$ field does not. The natural degree-23 action is faithful, so the same conclusion holds for their Galois closures: inertia in the closure of the unramified root field fixes every point, hence lies in the core of a point stabilizer, which is trivial.

The local analysis and specialization theorem answer complementary questions. The branch fibre reveals the centralizer and residue arithmetic after a $2A$ collision; the progression gives smooth arithmetic fibres with the full generic group.

10. Open questions

The osculating criterion narrows the fixed-point question to an explicit geometric existence problem.

1. Global uniqueness without the seven-model calculation. Can a global intersection argument show that

$$\mathcal{Z} = \mathcal{X} \cap H_A \cap H_B$$

is supported over exactly one geometric Hurwitz point, without testing all seven models? The local construction in theorem 1.4 supplies existence. The exclusion on the other six, and therefore global uniqueness, still comes from exact equations. An intersection-theoretic or moduli-theoretic proof of this uniqueness would strengthen the explanation of the fixed point.

2. Intrinsic construction and classification of the reductions. Can the explicit tail covers and the finite jet computations in section 5 be replaced by an intrinsic construction from the branch datum? We derive the canonical model and marked jets for the special map constructed there; they are no longer additional hypotheses in that application. We do not classify all special M_{23}-maps with the passport, or show that a harmonic cross-ratio by itself forces the same lift. A classification would explain which reductions can occur on the six-point component as well as why the constructed harmonic example is exceptional. A direct incidence relation in the deformation ring would also be preferable to recognizing an exact model and invoking uniqueness, provided it retains the effective critical divisor and the common sheet representation.

3. Finite incidence and other Nielsen classes. Can the finite Fano–affine incidence be extended relatively so that its specialization computes e_{sing}? The individual multiplicities and a compatible identification of the sheets must be retained. The calculation in section 6.3 explains why ordinary nearby-cycle duality [24], or logarithmic nearby cycles alone [21], do not identify the two prescribed invariants.

More generally, for other Nielsen classes with two totally ramified points, which canonical jet conditions or torsion-difference conditions cut out Galois-stable components? The genus-4 lemma here is elementary; finding an incidence whose support has the required size is the substantive part of such an application. The local Frobenius criterion alone cannot replace it.

4. Arithmetic of the branch-fibre tower. Classify the $2^4{:}\,\mathrm{PSL}_3(2)$-extensions of $\mathbf{Q}$ unramified outside $\{2, 3, 23\}$ with the constituent fields and discriminants of theorem 8.14. In particular, is the extension M unique?

11. Exact data and certificates

The repository accompanying this paper is

`https://github.com/research-reflexivity/frontier-math-m23-cover-investigation`.

It contains canonical JSON tables for the seven curves and marked points, the degree-23 maps, and the separate minimal equation P with its pencil (J_0, J_1). The osculating calculation uses the canonical models and marked points directly; it does not use the large minimal equation. The convention for the minimal-degree equation is

$$\texttt{coefficients\_T\_then\_W[i][j]} = [T^i W^j]P.$$

The polynomial has 120 nonzero coefficients and content 1; its coefficient of T^4W^{23} is

$$838610528294018704285119511337682389342017079161221371107109332287875.$$

The principal input fingerprints are

`data/Fint_coefficients_Z.json`
`27dd1fd0de4f1c350f5a07ead6d5b747d7a8f34e4146ec42fa953f1536a65102`

```
data/optimal_23_4_Z.json
cfb4185b2800744d524d87b3f08f5459cd38d42afb3787c11fa1e2aa3f25ee84
data/optimal_degree4_pencil.json
fdf571e4f47effc527e1726bdd99472be2c5e3390e45133fb3d7a3aab592619d
data/canonical_quadric_Q.json
352b978369353a1cb57a1a3f54edd6b7c024896edf78c38c1601a06f395ac943
data/hurwitz_algebra_candidate.json
10052c4ccace3cbeae0af7a307f01b6fc38c0b07404e99dd457c698fba07538a
data/hurwitz_canonical_models_candidate.json
f1594e9e85f3b19927938be1fa391afdb8f4ccc5a3db3ea41a2b2ab426a8e2c4
data/hurwitz_marked_points_candidate.json
129fb6c72262ddd14b18dd034d94f662cdf36de7e0eb432a725dcfc1283f0185
data/hurwitz_degree23_branch_candidate.json
cfbce099e3fa867b7c5ba29aa73f02f11d386cf60b00a506d1166d4c65639f62
data/hurwitz_degree23_maps_candidate.json
86cdbd314d01a9123aed0b6890fa4316b4303782b1d69563e020b6ede62a6c2a
```

The single command

```
make verify-all
```

runs the default open-source suite. The optional target `make verify-magma` repeats the minimal-model identity, irreducibility, canonical-quadric, and unpointed 23-inertia obstruction checks in Magma, together with the twelve-embedding Hurwitz critical-fibre and sextic Galois-closure checks. The recorded environment uses SageMath 10.9 [27], GAP 4.14.0 [28], Singular 4.4.1 [29], and PARI/GP 2.17.4 [30]; the independent Magma records use version 2.29-9 [26]. The opt-in target `make verify-hurwitz-third-fiber-exact` recomputes the longer characteristic-zero resultant certificate. The similarly opt-in target `make verify-hurwitz-connector-a6` reconstructs the formal annulus associated with the ramified A_6 calculation. This local calculation does not construct a semistable Galois closure or the missing relative incidence correspondences. The target `make verify-hurwitz-osculating` recomputes the canonical jets and homogeneous gcds over both coefficient fields, the residual-point divisor identity, and the quartic discriminant. It then repeats the incidence calculation after an invertible projective coordinate change. This is a fresh exact computation, not a check of a stored 2 versus 0 result. The target `make verify-hurwitz-degree-one-normalization` recomputes the third branch scaling whose residue changes the stored singular coordinate 16 to the normalized value -1. The new osculating and branch-normalization checks use SageMath; the existing Magma records do not certify these new results.

For theorem 5.9, the target `make verify-harmonic-reconstruction` checks the small exact model, smoothness, quintic sections, the full three-point passport, explicit descent to $\mathbf{Q}$, and the marked comparison modulo π^{33}. It then verifies an exact isomorphism with the earlier certified map, transferring its M_{23} monodromy certificate without a new Arb run. The separate, longer target `make audit-harmonic-hensel` repeats all 78 residual identity tests, the unit-Jacobian calculation, and an off-locus check which confirms that a residual incidence identity must not be treated as an exact identity. These are opt-in targets, separate from `verify-all`. The file `HARMONIC_RECONSTRUCTION.md` records their dependencies and the precise limits of the geometric conclusion.

The construction in section 5 has additional inputs. The tail function-field Galois computations are

```
notes/compute_rational_e8_tail_function_field_galois.m
notes/compute_q0_tail_function_field_galois.m.
```

Their separate summary records report successful Magma executions for the equations in equation (1). The open-source target `make audit-harmonic-construction` checks the deformation datum, tail ramification, direct symmetry calculations, coefficient valuations, first and higher jets, and the common-sheet Witt-design test. It checks the stored Magma input hashes, but does not rerun those two function-field computations. The node lemma, marked descent, and effective-divisor argument are written proofs, not conclusions of the polynomial tests.

The opt-in target `make verify-harmonic-magma` prepares and runs an independent Magma check of the exact model, base divisors, cross-products, and critical algebra. At this revision that new Magma check is prepared but unrun; it is not covered by the older Magma records. Neither generating its input nor verifying its checksum is counted as a successful Magma execution.

The target `make verify-hurwitz-frobenius-selector` checks the local decomposition and the finite-field identities in corollary 6.4.

There are different kinds of verification. The quick map verifier checks serialized coefficients and recorded ranks; the producer `scripts/reconstruct_hurwitz_degree23_maps.py`, run with `--component all --check-basepoints`, recomputes the multiplier systems and the residual base-locus saturations used in proposition 3.4. The branch-cycle summary verifier checks recorded permutations, their generated groups, Nielsen matching, and input hashes. Replaying the interval continuation requires the separate targets `certify-hurwitz-branch-cycles` and `certify-degree-one-branch-cycles`; a successful summary check is not a replay of all interval inequalities. Finally, finite group audits do not certify the written geometric arguments or construct the Fano–affine specialization left open in section 10.

The main targets are:

Result	Certificate	Engine
Global osculating incidence, residual points, and S_4 pencil	`verify_hurwitz_osculating.py`	SageMath
Degree-one third branch scaling	`verify_hurwitz_degree_one_branch_normalization.py`	SageMath
Newton-polygon valuations, nodes, adjoint pencil	`verify_boundary_valuations.gp`, `verify_nodes_mod31.sing`, `verify_adjoint_mod31.py`	PARI, Singular, Python
Minimal equation and function-field identity	`verify_optimal_23_4.py`, `verify_sage.py`, `verify_optimal_23_4.m`	Singular, SageMath, Magma
Specialization progression and ramification	`verify_progression_319.*`, `verify_ramified_t3830.*`	Python, PARI
Transport of the fibre over $T = 0$	`verify_special_fibre_bridge.py`	SageMath
Branch cycles and Fano incidence	`certify_branch_cycle_description.*`, `certify_special_fibre_fano_*`	GAP, PARI
Affine field tower and final quadratic layer	`certify_special_fibre_field_tower.*`, `certify_vanishing_cycle_class.*`	GAP, PARI
Ramification, flags, and complex-conjugation comparison	`certify_special_fibre_ramification.*`, `certify_local_flag_places.*`, `certify_real_frame_cocycle.*`	GAP, PARI
Geometric reflection obstruction and complex-conjugation comparison	`certify_geometric_reflection_*`	GAP, Magma

Result	Certificate	Engine
Canonical quadric and ruling field	`verify_canonical_quadric.py`, `verify_canonical_quadric.m`	SageMath, Magma
Unpointed 23-inertia obstruction	`certify_gauss_prolongation_obstruction.g`, `certify_gauss_prolongation_obstruction.m`	GAP, Magma
Coordinate algebra of the Hurwitz scheme, canonical curves, and marked points	`verify_hurwitz_*_candidate.py`	SageMath
Uniform atlas cover and truncation-error bounds	`verify_hurwitz_tail_geometry.py`, `verify_hurwitz_tail_summary.py`	Arb, Acb, Python
Degree-23 ratios, third branch value, and complete passport	`verify_hurwitz_degree23_*.py`, `verify_hurwitz_degree23_third_fiber.sage`, `verify_hurwitz_degree23_branch.m`, `verify_hurwitz_degree23_geometry.m`	SageMath, Magma
Exact degree-23 eliminant and all seven branch-cycle triples	`verify_hurwitz_monodromy_eliminant_candidate.py`, `verify_hurwitz_branch_cycle_summary.py`	SageMath, Arb
Transporter invariants and component-idempotent descent	`certify_hurwitz_relative_transporter.g`, `verify_hurwitz_relative_transporter.py`	GAP, SageMath
Galois closure and six-point permutation action	`verify_hurwitz_galois_closure.*`	SageMath, PARI, Magma
Local decomposition group and pointed reductions at 23	`verify_hurwitz_local_23.py`, `verify_hurwitz_pointed_23.py`	SageMath
Frobenius and quadratic-splitting idempotent	`verify_hurwitz_frobenius_selector.py`	SageMath
Finite identities and local obstructions to a specialization comparison	`notes/certify_ade_gluing_marker.py`, `notes/certify_wild_parameter_orientation.py`, `notes/audit_lifted_trace_fano_affine_incidence.g`, `notes/audit_fano_affine_odd_fixed_point_lemma.g`, `notes/audit_raw_lifted_trace_node_boundary.g`, `notes/audit_returned_normalizer_trace_parity.g`, `notes/explore_tagged_tame_boundary.g`, `notes/certify_pinched_tag_finite_identities.py`, `notes/audit_pointed_relative_bockstein.py`, `notes/audit_pointed_relative_bockstein.m`, `notes/audit_log_quadratic_orientation_line.py`, `notes/audit_log_quadratic_orientation_line.m`	GAP, Python, Magma

Several certificate filenames predate the terminology adopted in this paper and retain exploratory strings such as `connector`, `tagged`, or `orientation`. These are filenames, not names for additional geometric structures. Likewise, `special_fibre` in those filenames refers to the characteristic-zero fibre over $T = 0$, not to the closed fibre of a model over a local base.

For the coordinate-transport certificate, SageMath reconstructs the fibres over $T = 0$ in both plane models, checks the factor multiplicities and scalar identities, proves $\gcd(J_1(0, V), H_7R_8) = 1$, and computes the minimal polynomial of J_0/J_1 in each residue field. Thus the certificate verifies the coordinate transport coefficient by coefficient, including regularity and residue-field generation.

The largest computations in the Fano suite are the direct degree-1344 splitting-field calculation for R_8, the comparison of the possible septic/octic Frobenius types, the 9,889 transvection primes below 10^6, and the square-class tests at fourteen primes splitting completely in L_0.

12. Scope, provenance, and disclosure

The starting regular M_{23} realization, its source equation, branch cycles, and the numerical computation of the seven complex covers and their Galois-fixed Nielsen class are due to Huang, Jackson, Lee, Poonen, Pries, and Zhang [1]. Their accompanying repository supplies the defining data used here [2]. The local reconstruction in section 5.6 recovers the same cover; it is not a claim to a new regular realization.

The contributions of the present paper fall into five groups.

(1) We reconstruct the finite inner Hurwitz scheme, its seven canonical curves and degree-23 maps, and their certified branch cycles, and determine the S_6 Galois action on the sextic component.
(2) We construct the relative osculating incidence and prove by exact calculation that it has nonempty fibre on exactly one cover. Its functoriality gives a global Galois-fixedness argument. We prove its equivalent torsion-difference criterion and determine the S_4 monodromy of the associated degree-4 pencil.
(3) We determine the local decomposition and pointed reductions at 23, including the uniformly normalized harmonic position. We compare the component idempotents obtained from osculating incidence, singular position, and finite-group augmentation. The local idempotent also describes a Frobenius-fixed complement and an unramified quadratic-splitting locus. We construct a harmonic special M_{23}-map from explicit tail covers, derive the canonical model and required marked jets of its lift, and prove incidence by Hensel uniqueness and an independently verified small exact model. A separate exact isomorphism identifies that model with the certified cover.
(4) On the distinguished cover, we explicitly construct the separate degree-4 coordinate anticipated in [1, Remark 3.4], its bidegree-$(23, 4)$ equation, and its branch-fibre arithmetic: the Fano plane, affine torsor, field tower, ramification, and flags.
(5) We prove the uniform specialization theorem and controlled ramification inside its progression, and choose an infinite mutually independent family there. The abstract existence of such families is already [1, Corollary 1.4(d)].

The global incidence argument does not define the distinguished point by its known rationality. It defines a Galois-invariant geometric condition and uses exact models to verify that exactly one point satisfies it. The local construction gives a separate existence argument, starting with characteristic-23 equations rather than the seven generic models. It is also computer-assisted: the function-field groups, the finite jet equations, and the small exact model are verified algebraic inputs. The passage between them uses the written lifting, descent, canonical-lattice, effective-divisor, and node arguments of section 5. Those geometric arguments require mathematical review beyond the successful execution of the algebraic checks.

For this constructed special map the model and marked jets are derived, not assumed. We do not claim an equation-free deduction from the branch classes, a classification of all special maps with this passport, or a general implication from harmonic reduction to incidence. Global fixedness uses the intrinsic incidence and its single-point support; a local rank-one factor alone would not give that conclusion. The explicit rational descent of the small model is an additional exact check, not the premise defining the incidence locus.

The relative scheme $\mathcal{Z}$ is a characteristic-zero construction on the universal quotient curve. It is not the previously proposed relative Fano–affine correspondence, and it does not supply the latter's specialization. The component comparison still uses the exact model and branch-cycle identifications. No relative quadratic determinant line or Fano–affine specialization proof is asserted. The elementary nearby-cycle stalk retains the individual sheets; the unrestricted coefficient Bockstein does not recover the specified integral quadratic invariant.

The third branch fibre of the six conjugate maps is verified exactly in characteristic zero and independently in all twelve residue embeddings of a completely split prime. This distinction matters: numerical and lattice reconstruction discover the coefficients, while the subsequent exact elimination and branch-cycle certificates identify the covers.

The discovery and drafting process used substantial AI assistance. The computational evidence and the scope of each verification are described in section 11. The new generic-model Magma check remains unrun; the older tail function-field records are identified separately. None of these records certifies the written geometric reasoning or supplies the unresolved Fano–affine relative construction.

Acknowledgements. This work analyzes the construction of Huang, Jackson, Lee, Poonen, Pries, and Zhang and depends on their making the defining data publicly available in the repository cited as [2].

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